Self-Mutual Information-Based Band Selection for Hyperspectral Image Classification

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***Abstract*— Due to significant inter-band correlation resulting from the use of hundreds of contiguous spectral bands, band selection (BS) is commonly used to reduce data dimensionality for band redundancy removal. A challenge for BS is how to design an effective criterion which can select bands with crucial self-retained spectral information, while also avoiding highly correlated bands to be selected. This article presents a novel approach, referred to as self-mutual information-based band selection (SMI-BS) for hyperspectral image classification (HSIC) to address these two issues. It first constructs a hyperspectral band channel from a hyperspectral image and then takes advantage of such a band channel to coin a new concept of SMI, which is defined as the mutual information (MI) between a selected band, b, and the set of full bands, Ω, $I(\mathbf{b};\Omega)$. As a result, a curve plotted as a function of $I(\mathbf{b};\Omega)$ versus individual band b, called SMI curve, can be used as a BS criterion which selects those bands with large $I(\mathbf{b};\Omega)$ values as desired bands. Since such selected bands may be highly correlated, another new concept, called prominent band (PB), which is defined as a band corresponding to a prominent peak of an SMI curve, is further introduced to avoid selecting highly inter-correlated spectral bands. To validate the utility of SMI-BS in HSIC, experiments are conducted to compare existing state-of-the-art BS methods for performance evaluation. The results demonstrate that SMI-BS is indeed a very effective BS method and also performs better than other test BS methods.**

***Index Terms*— Mutual information (MI), prominent band (PB), self-mutual information (SMI), virtual dimensionality (VD).**

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I. Introduction

INFORMATION theory [1]–[3] has recently found its potential in hyperspectral data exploitation and has been also applied to various areas such as spectral information divergence (SID) developed to measure spectral variability [4], [5], channel capacity (CC)-based band subset selection (BSS) for band selection (BS) [6], interband spectral capacity [7], sample entropy and class entropy in [8], and class information used to selected bands for hyperspectral image classification (HSIC) [9]. This article develops a new information theoretic approach to BS, to be called self-mutual information (SMI)-based BS (SMI-BS). It first defines a new concept of SMI, denoted by $I(\mathbf{b};\Omega_Y)$, as the mutual information (MI) between one single band, $\mathbf{b}$ and the full band set, Ω_Y as the band channel output space Y, and then uses SMI as a band prioritization (BP) criterion to rank all bands for BS. This proposed SMI-based BS is quite different from the CC-based BSS (CC-BSS) developed in [6], which uses CC as a BS criterion to select band subsets. Specifically, SMI-BS considers the MI between a particular band of interest and the full band set Ω rather than MI between the entire channel input space and channel output space. The role of SMI plays in MI is similar to self-information that defines entropy [10].

A channel is composed of three components, channel input space, Ω_X, channel output space Ω_Y, and a channel transition matrix $\mathbf{Q} = [Q_{y|x}]_{x\in\Omega_X, y\in\Omega_Y}$ as a triplet (Ω_X,Ω_Y,$\mathbf{Q}$). In order to apply the concept of a channel to hyperspectral imaging, we assume that a hyperspectral image cube is represented by L spectral bands, $\{\mathbf{b}_l\}_{l=1}^{L}$. Then, a hyperspectral band channel can be constructed by letting the hyperspectral band channel input space and the hyperspectral band channel output space be specified by $\mathbf{\Omega}_X = \{\mathbf{b}_j\}_{j=1}^{L}$ and $\mathbf{\Omega}_Y = \{\mathbf{b}_k\}_{k=1}^{L}$, respectively, and the channel transition matrix be specified by an $L \times L$ conditional transition probabilities, $\mathbf{Q} = [Q_{k|j}]_{j=1,k=1}^{L,L}$ which is a matrix formed by a set of conditional probabilities $Q_{k|j} = p(\mathbf{b}_k|\mathbf{b}_j)$ that describe a transition from the jth band, $\mathbf{b}_j$ in Ω_X to the kth band, $\mathbf{b}_k$ in Ω_Y. Then, a hyperspectral band channel is characterized by a triplet ($\mathbf{\Omega}_X = \{\mathbf{b}_j\}_{j=1}^{L}, \mathbf{\Omega}_Y = \{\mathbf{b}_k\}_{k=1}^{L}, \mathbf{Q} = [Q_{k|j}]_{j=1,k=1}^{L,L}$). The only remaining issue is how to construct $\mathbf{Q} = [Q_{k|j}]_{j=1,k=1}^{L,L}$. An approach developed in [5] and [6] is to use a band discrimination (BD) measure for this purpose. By virtue of such a constructed hyperspectral band channel, the MI between the band channel input space $\mathbf{\Omega}_X$ and

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the band channel output space Ω_Y, denoted by $I(\Omega_X; \Omega_Y)$, can be well-defined according to [1]–[3].

One of fundamental theorems in information theory is Shannon's Second coding theorem [1]–[3] which states that an information source X can be transmitted reliably with error probability asymptotically approaching zero with the information rate, R, no less than the source entropy, $H(X)$ and no greater than CC, C, i.e., $H(X) \leq R \leq C$ where CC is the channel capacity defined as the maximal MI over all possible source input probability distributions. By taking advantage of CC, Chang *et al.* [6] developed a CC-based approach to BS, called channel capacity-based band subset selection (CC-BSS), to select an optimal band subset.

This article takes a rather different view from CC used in [6] to develop a new BS approach which has never been explored in the literature. Its idea explores insights into how CC is defined by SMI in a similar manner that how the entropy is defined by self-information [10]. In other words, what self-information is to entropy in the sense that the entropy is the mean of self-information with respect to a source input provability distribution can be also done for what SMI is to MI in the sense that MI is the mean of SMI with respect to a source input probability distribution. In fact, self-information was investigated in [8] to define class significance to determine the number of required bands to be selected for each class for HSIC. Inspired by self-information, SMI is also developed similarly to calculate how much MI of a full band set Ω can provide for an individual band and then uses it as *a priori*ty criterion to rank this particular band for BS. However, this is easily said than done since self-information is defined by a single information source compared with SMI which is defined by a channel between two sources, a band channel input and band channel output space. So, what is done for self-information cannot be duplicated and directly applied to SMI. Accordingly, a new mechanism must be sought.

Fortunately, Theorem 4.5.1 proved in [2] states that the SMI $I(\mathbf{b};\Omega_Y)$ between any band $\mathbf{b}$ and Ω is right between $I(\Omega_X;\Omega_Y)$, which is the mean of $I(\mathbf{b};\Omega_Y)$ with $\mathbf{b}$ averaged over the band channel output space Ω_Y and CC which is the maximum MI over all possible input distribution. When a source input probability distribution achieves CC, the $I(\mathbf{b};\Omega_Y) \leq$ CC with $I(\mathbf{b};\Omega_Y) =$ CC for any band $\mathbf{b}$ with its nonzero probability. This theorem lays the foundation for developing SMI as a BS criterion to prioritize a band $\mathbf{b}$.

Despite the fact that entropy and relative entropy [also known as directed information divergence (ID)] have been used as BP criteria to rank all spectral bands, their ideas are focused on how to measure the entropy of each individual band rather than the correlation among bands. SMI offers a new look into the BS design criterion by describing the correlation between a single band and full bands via SMI through a constructed hyperspectral band channel so that band correlation arising in BP-based BS methods can be resolved by SMI.

To have a better logic to present the proposed SMI-BS approach, this article is organized as follows. Section II reviews related works and summarizes novelties of the proposed SMI-BS approach. Section III is devoted to how to construct a hyperspectral band channel. Section IV reinterprets MI by SMI so that CC can be reexpressed as the mean of SMI over all channel input signal sources. Section V describes the SMI curve and finds its prominent bands (PBs). Section VI details the implementation of the SMI-BS algorithm. Section VII includes images to be studied and test BS methods along with classification measures to be used to conduct a comparative analysis. Section VIII discusses experimental results. Section IX provides computing time complexity analysis. Finally, Section X draws some conclusions.

II. Related Works and Novelties of SMI-BS

In the following, we review some related works to SMI-BS and also summarize several new and novel ideas.

1) The most important novel idea is the design of SMI to measure spectral correlation between a single selected band and a full band set via a hyperspectral band channel. Although modeling a hyperspectral image as a hyperspectral band channel was not new [6], [7], the proposed SMI-BS takes advantage of SMI to offer an alternative new approach. Instead of using CC as a criterion, SMI-BS uses the input probability distribution found by Blahut's algorithm for CC [11] to calculate SMI. It follows a similar idea to the concept of self-information [10] used to define entropy to define SMI as a BP criterion to select bands. Compared with CC-BSS in [6] which selects a desired band subset between two band channel input and output spaces, SMI-BS selects bands according to SMI between a selected band and a full band set. More specifically, SMI-BS prioritizes all bands, while CC-BSS uses band search strategies to select band subsets. So, SMI-BS is conceptually different from the CC-BSS.
2) Despite that SMI-BS is a BP-based BS, it distinguishes itself from many existing BP-based methods, such as variance, signal-to-noise ratio (SNR), entropy, ID [12], and Jeffries–Matusita (JM) distance [13] in the sense that these methods are single band-based BP criteria as opposed to SMI-BS which is a band-to-band subset criterion. In particular, SMI takes advantage of a hyperspectral band channel to prioritize bands, while finding PBs to avoid selecting high-correlated bands, a task which a BP-based method generally cannot accomplish.
3) As an alternative to BP-based BS methods, band clustering-based methods have also been investigated. For example, Wang *et al.* [14] developed an adaptive subspace partition strategy (ASPS)-based BS method by partitioning a hyperspectral image cube into multiple subcubes via maximizing the ratio of interclass distance to intraclass distance. Most recently, Wang *et al.* [15] further developed a fast neighborhood grouping method for hyperspectral band selection (FNGBS) which partitions a hyperspectral image cube into several groups using *coarse-fine* strategy so that the bands with the maximum product of local density and information entropy in groups are selected as a band subset. However, FNGBS used information entropy not MI as a criterion for BS to measure individual bands. Most recently, Baisantry *et al.* [16] developed an interesting feature extraction-based, clustering-ranking type BS method

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which uses a sparse subspace clustering to cluster similar bands into groups and prioritized bands by a new metric, called combined divergence-correlation index to select the most discriminative as well as the least correlated bands as cluster representatives. Compared with the abovementioned methods, SMI-BS does not partition a hyperspectral image cube into a finite number of subcubes or groups by different partition strategies as proposed in [14]–[16], but rather calculates SMI to measure the correlation between a selected band and the full band set to prioritize each selected bands.

4) In order to avoid selecting highly correlated bands, BSS-based methods must rely on various search strategies. For example, Chang *et al.* [6] derived CC-BSS which used sequential (SQ) and successive (SC) algorithms developed for endmember extraction in [17]–[23] to select band subsets. There are also other methods devoted to BSS, such as particle swarm optimization (PSO) in [24], firefly algorithm (FA) in [25], multitask sparsity pursuit (MTSP) [26], multigraph determinantal point process (MDPP) [27], and dominant set extraction BS (DSEBS) in 28]. Unfortunately, since exhausting all possible band subset combinations is practically infeasible, most BSS-based methods actually only run a small number of selected candidates of band subsets to find suboptimal band subsets.

   As a complete opposite to the abovementioned BSS methods, SMI-BS provides a simple means of solving this problem. SMI-BS plots the SMI curve of all bands and then uses the concept of prominent peaks to select desired bands as PBs, a simple novel idea which resolves the issue of interband correlation and has never been explored in the literature. In other words, SMI-BS coupled with PB-finding can indeed achieve what BSS-based methods can accomplish. This is a tremendous advantage over BSS-based methods in the sense that the latter generally produces suboptimal band subsets because it does not run through all possible band subset combinations.

5) Finally, one novelty of SMI is noteworthy. The concept of MI has been widely used in communications/signal processing, particularly in image registration. It can be reexpressed as the average of SMI over all input signal sources. This new interpretation may open up new directions of looking into many MI-based works such as image registration not necessarily limited to BS proposed in this article.

III. Construction of a Hyperspectral Band Channel

The key component of a hyperspectral band channel is the construction of a channel transmission matrix. Assume that $\{\mathbf{b}_l\}_{l=1}^{L}$ is a total set of spectral band images representing a hyperspectral image cube where $\mathbf{b}_l$ is the lth spectral band image represented as a column vector, $\mathbf{b}_l = (b_{l1}, b_{l2}, \ldots, b_{lN})^T$, and N is the total number of data samples in the band image. Now, we can define a band channel $(\mathbf{\Omega}_X, \mathbf{\Omega}_Y, \mathbf{Q} = [Q_{k|j}]_{j=1,k=1}^{L,L})$ shown in Fig. 1 where the channel input space, the channel output space by

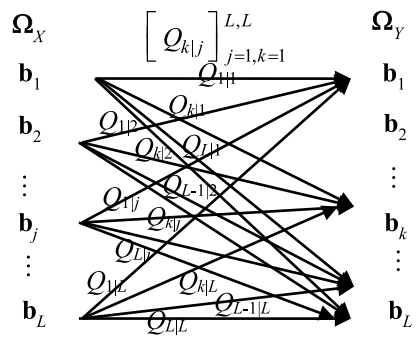


Fig. 1. Hyperspectral band channel specified by $\mathbf{\Omega}_X = \mathbf{\Omega}_Y = \{\mathbf{b}_l\}_{l=1}^{L}$.

$\mathbf{\Omega}_Y = \{\mathbf{b}_l\}_{l=1}^{L}$, and the channel transition probabilities are specified by $\mathbf{\Omega}_X = \{\mathbf{b}_l\}_{l=1}^{L}$, $\mathbf{\Omega}_Y = \{\mathbf{b}_l\}_{l=1}^{L}$, and

$$Q_{k|j} = \mathrm{BD}(\mathbf{b}_k, \mathbf{b}_j) \Big/ \sum_{k=1}^{L} \mathrm{BD}(\mathbf{b}_k, \mathbf{b}_j) \tag{1}$$

respectively, with BD defined by an arbitrary spectral BD measure [4], [5], [11].

In this article, three BD measures, SID in [4], [5], and [29], spectral angle mapper (SAM) [29], and SIDAM in [5] will be used to construct band channel, each of which is described in detail as follows.

A. SID Used to Construct a Hyperspectral Band Channel

SID was originally developed in [4] to measure the discrepancy between two spectral bands based on their probability distributions. To calculate SID, we first define the probability vector $\mathbf{p}_i = (p_{i1}, p_{i2}, \ldots, p_{iN})^T$ for band $\mathbf{b}_i$ and $\mathbf{p}_k = (p_{k1}, p_{l2}, \ldots, p_{lN})^T$ for $\mathbf{b}_k$ as follows:

$$p_{il} = b_{il} \Big/ \sum_{n=1}^{N} b_{\mathrm{in}} \quad \text{and} \quad p_{kl} = b_{kl} \Big/ \sum_{n=1}^{N} b_{kn}. \tag{2}$$

Then, SID between two bands $\mathbf{b}_j$, $\mathbf{b}_k$ is defined by

$$\mathrm{SID}(\mathbf{b}_j, \mathbf{b}_k) = D(\mathbf{p}_j || \mathbf{p}_k) + D(\mathbf{p}_k || \mathbf{p}_j) \tag{3}$$

where

$$D(\mathbf{p}_j || \mathbf{p}_k) = \sum_{n=1}^{N} p_{\mathrm{in}} \log \frac{p_{\mathrm{in}}}{p_{kn}}. \tag{4}$$

By (3), the channel transition probability can be defined as

$$Q_{k|j} = \mathrm{SID}(\mathbf{p}_k, \mathbf{p}_j) \Big/ \sum_{k=1}^{L} \mathrm{SID}(\mathbf{p}_k, \mathbf{p}_j). \tag{5}$$

B. SAM Used to Construct a Hyperspectral Band Channel

SAM is another popular spectral measure defined by

$$\mathrm{SAM}(\mathbf{b}_j, \mathbf{b}_k) = \cos^{-1}\left(\frac{\mathbf{b}_j^T \mathbf{b}_k}{\|\mathbf{b}_j\| \|\mathbf{b}_k\|} \right). \tag{6}$$

where $\|\mathbf{b}\|$ is defined as the length of $\mathbf{b}$. Then, the channel transition probability can be defined as

$$Q_{k|j} = \mathrm{SAM}(\mathbf{b}_k, \mathbf{b}_j) / \sum_{k=1}^{L} \mathrm{SAM}(\mathbf{p}_k, \mathbf{p}_j). \tag{7}$$

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C. SIDAM Used to Construct a Hyperspectral Band Channel

A third measure is SIDAM developed by Du *et al.* [5], which combines SID and SAM as follows:

$$\mathrm{SIDAM}(\mathbf{b}_j, \mathbf{b}_k) = \mathrm{SID}(\mathbf{b}_j, \mathbf{b}_k) \tan\big(\mathrm{SAM}(\mathbf{b}_j, \mathbf{b}_k)\big). \quad (8)$$

IV. Reinterpretation of MI by SMI

From information theory in [1]–[3], the entropy of a band image $\mathbf{b}_i$ specified by its probability distribution, $\mathbf{p}_i = (p_{i1}, p_{i2}, \ldots, p_{iN})^T$, is defined by

$$H(\mathbf{p}_i) = -\sum_{n=1}^{N} p_{\mathrm{in}} \log p_{\mathrm{in}}. \quad (9)$$

However, according to [10], we can define the self-information of the nth data sample in the ith band image, r_{in} as

$$I(r_{\mathrm{in}}) = -\log p_{\mathrm{in}}. \quad (10)$$

Using (10), the entropy in (9) can be reexpressed as

$$H(\mathbf{p}_i) = \sum_{n=1}^{N} p_{\mathrm{in}} I(r_{\mathrm{in}}) \quad (11)$$

which is the mean of self-information over all the data samples in the band image $\mathbf{b}_i$.

Inspiring by the reexpression of the entropy in (11), we can also look into the definition of MI in [1]–[3], and [11] and define a hyperspectral band channel in Fig. 1 by (1) as

$$I(\mathbf{\Omega}_X; \mathbf{\Omega}_Y) = \sum_{j=1}^{L} q_j \sum_{k=1}^{L} \log \frac{Q_{k|j}}{\sum_j^L q_j Q_{k|j}} \quad (12)$$

where the probability vector $\mathbf{q} = (q_1, q_2, \ldots, q_L)^T$ is a probability distribution defined on the channel input space $\mathbf{\Omega}_X = \{\mathbf{b}_j\}_{j=1}^L$ and q_j is the probability of the jth channel band input $\mathbf{b}_j$. It should be noted that the probability vector $\mathbf{q}$ is different from the probability vector $\mathbf{p}_i$ used to define entropy in (9).

Now, if we further define $I(\mathbf{b}_j; \mathbf{\Omega}_Y)$ as SMI between $\mathbf{b}_j$ and the full band set, Ω specified by the band channel output space $\mathbf{\Omega}_Y$ as

$$I(\mathbf{b}_j; \mathbf{\Omega}_Y) = \sum_{k=1}^{L} Q_{k|j} \log \frac{Q_{k|j}}{\sum_{j=1}^{L} q_j Q_{k|j}} \quad (13)$$

then the MI in (12) can be reexpressed in a similar manner that the entropy in (9) is reexpressed by (11) as

$$I(\mathbf{\Omega}_X; \mathbf{\Omega}_Y) = \sum_{j=1}^{L} q_j I(\mathbf{b}_j; \mathbf{\Omega}_Y). \quad (14)$$

Based on information theory [2], [3], the capacity of a hyperspectral band channel described in Fig. 1 is defined by the maximum MI over all possible channel band input probability vectors $\mathbf{q} = (q_1, q_2, \ldots, q_L)^T$ as

$$C = \max_{\mathbf{q}} I(\mathbf{\Omega}_X; \mathbf{\Omega}_Y). \quad (15)$$

Using (13)–(15), we can reinterpret the CC theorem described by Theorem 4.5.1 in [2, p. 91] or Theorem 2 in [11] as a CC theorem for a hyperspectral band channel as follows.

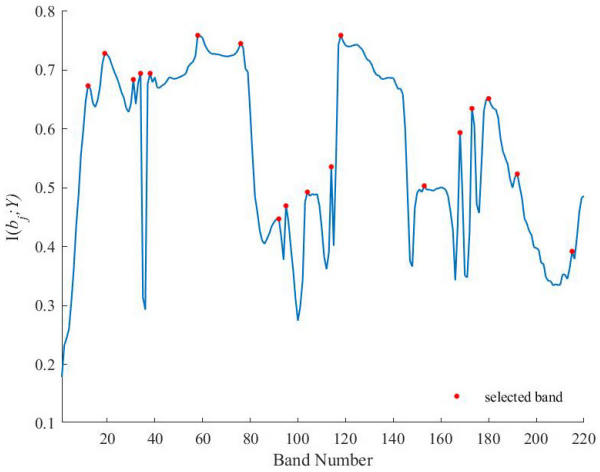


Fig. 2. Curve of $I(\mathbf{b}_j; \Omega_X)$ produced by SIDAM with 18 local maxima selected as bands for the Purdue data.

Theorem 1: Capacity for a Hyperspectral Band Channel

For any input probability distribution $\mathbf{q}$ defined on $\mathbf{\Omega}_X = \{\mathbf{b}_j\}_{j=1}^L$ that achieves CC of a discrete memoryless hyperspectral band channel $\{Q_{k|j}\}_{j=1,k=1}^{L,L}$ described by (1), a necessary and sufficient condition is

$$I(\mathbf{b}_j; \mathbf{\Omega}_Y) = C \text{ for all } j \text{ with } q_j = q(\mathbf{b}_j) > 0 \quad (16)$$

and

$$I(\mathbf{b}_j; \mathbf{\Omega}_Y) \le C \text{ for all } j \text{ with } q_j = q(\mathbf{b}_j) = 0 \quad (17)$$

where $\mathbf{\Omega}_Y = \{\mathbf{b}_j\}_{j=1}^L$. Specifically, the following inequalities also hold as shown in Theorem 5.4.7 [3, p.169]:

$$\sum_{j=1}^{L} q_j I(\mathbf{b}_j; \mathbf{\Omega}_Y) \le C \le \max_{1 \le j \le L} I(\mathbf{b}_j; \mathbf{\Omega}_Y). \quad (18)$$

By virtue of (16) and (17), $I(\mathbf{b}_j; \mathbf{\Omega}_Y)$ defined by (13) is SMI provided by the jth input band $\mathbf{b}_j$ with respect to the band channel output space $\mathbf{\Omega}_Y$ and achieves the CC when $I(\mathbf{b}_j; \mathbf{\Omega}_Y)$ is constant for all band images $\mathbf{b}_j$ with $q_j = q(\mathbf{b}_j) > 0$.

V. SMI Curve and Finding PBs

According to (18), the higher $I(\mathbf{b}_j; \mathbf{\Omega}_Y)$ is, the more information the band $\mathbf{b}_j$ possesses until it reaches CC. By using this property, $I(\mathbf{b}_j; \mathbf{\Omega}_Y)$ can be used as a quantitative measure as a BP criterion to rank a band with respect to the full band set Ω. However, the correlation among adjacent bands is likely to be high. To deal with this issue, we plot the curve of $I(\mathbf{b}_j; \mathbf{\Omega}_Y)$ versus $\mathbf{b}_j$ across the band channel input space Ω_X, defined as SMI curve. Then, a prominent peak of the SMI curve is used to measure how much the peak stands out due to its intrinsic height and its location relative to other peaks on the SMI curve. In this case, a PB can be specified by a prominent peak and

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Algorithm 1 Algorithm for Finding Prominent Bands

1. Calculate the prominence of $\mathbf{b}_{j_k} \in \mathbf{\Omega}_{\mathrm{LMX}}$, denoted by $\mathrm{prom}(\mathbf{b}_{j_k})$ by the following procedure

a. Extend a horizontal line from the point $\mathbf{b}_{j_k}$ to the left and the right until the line either

- crosses the curve because there is a higher point or
- reaches the left or right end of the curve

Then, the left interval and right interval of $\mathbf{b}_{j_k}$ are defined.

b. Find the minimum point in each of the two intervals defined in 1(a).

c. Use the higher point of the two interval minima as the reference level. The prominence is defined as the height of $\mathbf{b}_{j_k}$ above the reference level.

2. Then the peaks of the SMI curve are defined by $\Omega_{\mathrm{PK}} = \{\mathbf{b}_{j_l}\}_{l=1}^{n_{\mathrm{PK}}}$ where $\mathbf{b}_{j_l} \in \mathbf{\Omega}_{\mathrm{LMX}}$ and $\mathrm{prom}(\mathbf{b}_{j_l}) \geq 10^{-2}$. The threshold 10^{-2} is determined empirically by experiments.

those bands which yield local maxima defined as prominent peaks will be, therefore, selected as desired bands. Using the Purdue data in Section VII-A1 as an illustrative example, Fig. 2 shows the curve plotted for $I(\mathbf{b}_j; \mathbf{\Omega}_Y)$ versus $\mathbf{b}_j$ with $1 \leq j \leq L$ where the bands marked by red dots are prominent peaks selected for BS because they have the local maxima of $I(\mathbf{b}_j; \mathbf{\Omega}_Y)$.

Since a function of finding prominent peaks used in the SMI-BS is available from Mathworks' website, https://www.mathworks.com/help/signal/ref/findpeaks.html, its detailed descriptions are not included here but rather moved to the end of this article as an appendix for reference. According to the appendix, let the local maximum points set be denoted by $\mathbf{\Omega}_{\mathrm{LMX}} = \{\mathbf{b}_{j_k}\}_{k=1}^{n_{\mathrm{LMX}}}$ where n_{LMX} is the number of local maximum points.

VI. SMI-Based BS

In this section, an SMI-based BS algorithm is developed. Since it requires the number of bands to be selected, n_{BS}, the virtual dimensionality (VD) [29]–[31] is used to determine n_{BS}. However, an issue arises when the number of prominent peaks n_{PK} of the $I(\mathbf{b}_j; \mathbf{\Omega}_Y)$curve is less than n_{VD}. To resolve this issue, uniform band selection (UBS) is implemented in conjunction with n_{PK}. It divides the remaining bands into $(n_{\mathrm{VD}} - n_{\mathrm{PK}})$ intervals. Then, for each interval, the band with the largest SMI is selected. The detailed step-by-step implementation is given in the following algorithm.

Self-Mutual Information-Based Band Selection (SMI-BS)

Step 1: Initial condition: Let n_{VD} be the number of bands needed to be selected, n_{BS} determined by VD.

Step 2: Construct a band discrimination channel
- (i) Let the channel input space and channel output space specified by $\mathbf{\Omega}_X = \{\mathbf{b}_j\}_{j=1}^{L}$ and $\mathbf{\Omega}_Y = \{\mathbf{b}_k\}_{k=1}^{L}$, respectively.
- (ii) Construct a channel transition matrix $\mathbf{Q} = [Q_{k|j}]_{j=1,k=1}^{L,L}$ by SID, SAM, SIDAM.

Step 3: Implement the Blahut algorithm [11] to find an input probability distribution $\mathbf{q}^*$ which achieves CC and use $\mathbf{q}^*$ to further find the mutual information $I(\mathbf{b}_j; \mathbf{\Omega}_Y)$ between $\mathbf{b}_j$ and $\mathbf{\Omega}_Y$ to form a band mutual information set denoted by $\{I(\mathbf{b}_j; \mathbf{\Omega}_Y)\}_{j=1}^{L}$.

Step 4: Plot the curve of $\{I(\mathbf{b}_j; \mathbf{\Omega}_Y)\}_{j=1}^{L}$ versus $\{\mathbf{b}_j\}_{j=1}^{L}$, called SMI curve and find all bands which yield the prominent peaks of $I(\mathbf{b}_j; \mathbf{\Omega}_Y)$-generated curve, denoted by $\Omega_{\mathrm{PK}} = \{\mathbf{b}_{j_l}\}_{l=1}^{n_{\mathrm{PK}}}$ where n_{PK} be the number of prominent peaks, i.e., $n_{\mathrm{PK}} = |\Omega_{\mathrm{PK}}|$.

Step 5: Band selection
- (i) If $n_{\mathrm{PK}} \geq n_{\mathrm{VD}}$, the bands with the first n_{VD} highest values of $I(\mathbf{b}_j; \mathbf{\Omega}_Y)$ with $\mathbf{b}_j$ in $\mathbf{\Omega}_{\mathrm{PK}}$ are selected as desired bands, denoted by Ω_{BS}.
- (ii) If $n_{\mathrm{PK}} < n_{\mathrm{VD}}$, let $\tilde{\mathbf{\Omega}}_X = \mathbf{\Omega}_X - \mathbf{\Omega}_{\mathrm{PK}}$, $K = n_{\mathrm{VD}} - n_{\mathrm{PK}}$.
  - a. Uniformly divide $\tilde{\mathbf{\Omega}}_X$ into K intervals such that $\{\Delta_k\}_{k=1}^{K}$ where Δ_k is the kth band interval which contains $\Delta_k = \{\mathbf{b}_{k_i}\}_{i=1}^{n_k}$. For each Δ_k find a band $\mathbf{b}_k^*$ which has the maximum mutual information $\mathbf{b}_{k^*} = \arg\left\{\max_{\mathbf{b}\in\Delta_k} I(\mathbf{b}_j; \mathbf{\Omega}_Y)\right\}$, i.e., $I(\mathbf{b}_{k^*}; \mathbf{\Omega}_Y) = \max_{\mathbf{b}\in\Delta_k} I(\mathbf{b}_j; \mathbf{\Omega}_Y)$. Let $\tilde{\mathbf{\Omega}} = \{\mathbf{b}_k^*\}_{k=1}^{K}$
  - b. Final selected band subset is $\mathbf{\Omega}_{\mathrm{BS}} = \mathbf{\Omega}_{\mathrm{PK}} \cup \tilde{\mathbf{\Omega}}$.

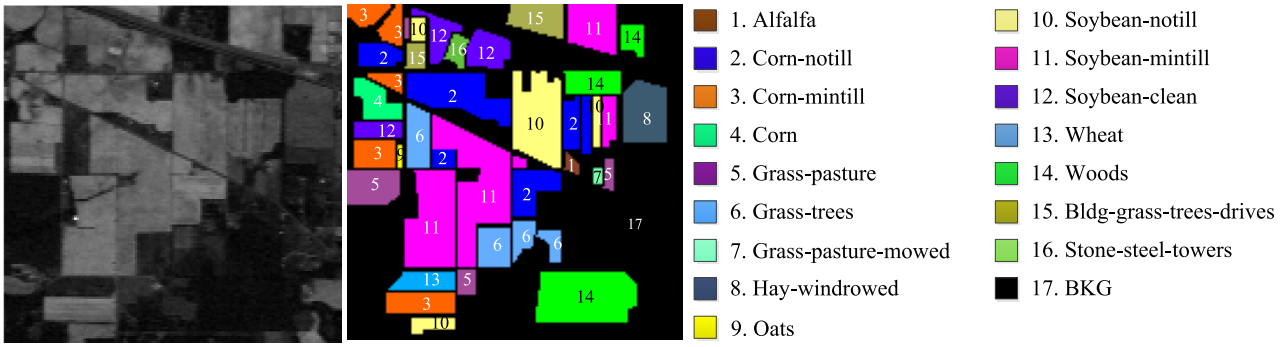
VII. Images and Test BS Methods Along With Classification Measures Used for Experiments

A. Hyperspectral Images Used for Experiments

Three popular hyperspectral images were used for experiments and also studied extensively in HSIC, each of which has its own merit to be studied for SMI-BS.

1) Purdue Indiana Indian Pines: The first one is the Purdue Indiana Indian Pines test site with an aerial view shown in Fig. 3(a) along with its ground truth of 17 class maps in Fig. 3(b) and their class labels in Fig. 3(c). Table I tabulates the number of data samples in each class where there are four small classes with less than ten samples; classes 7, 9, 1, 16, and 3 classes with more than 1000 samples; classes 14, 2, and 11. So, this scene clearly has an imbalanced

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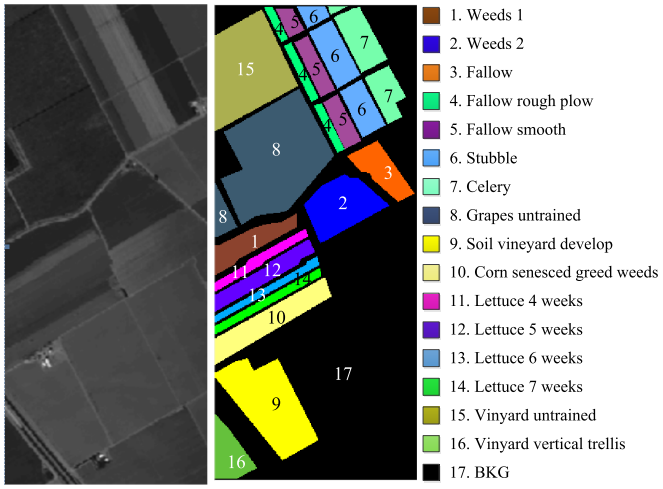


(a) (b) (c)

Fig. 3. Purdue's data and its ground truth with 16 classes. (a) Band 186 (2162.56 nm). (b) Ground truth map. (c) Classes by colors.

TABLE I
CLASS LABELS OF PURDUE'S DATA

class 1 (46)	Alfalfa	class 7 (28)	grass/pasture-mowed	class 13 (205)	wheat
class 2 (1428)	corn-notill	class 8 (478)	hay-windrowed	class 14 (1265)	woods
class 3 (830)	corn-min	class 9 (20)	oats	class 15 (386)	Bldg-grass green-drives
class 4 (237)	corn	class 10 (972)	soybeans-notill	class 16 (93)	stone-steel towers
class 5 (483)	grass/pasture	class 11 (2455)	soybeans-min	class 17 (10249)	BKG
class 6 (730)	grass/trees	class 12 (593)	soybeans-clean		



(a) (b) (c)

Fig. 4. Salinas and its ground truth with 16 classes. (a) Salinas scene. (b) Ground-truth image. (c) Classes by colors.

class issue in classification. It is an Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) image scene and has size of $145 \times 145 \times 220$ pixel vectors with water absorption bands (bands 104–108 and 150–163, 220). So, a total of 220 bands were used for experiments. It should be noted that in many reports, 200 bands were used by excluding water absorption bands such as [32]. However, for BS, it is believed that a full set of 220 band should be used for band integrity.

2) Salinas: A second image data set is Salinas in Fig. 4(a) which is also an AVIRIS scene. It was collected over Salinas Valley, California with a spatial resolution of 3.7-m per pixel with spectral resolution of 10 nm. It has size of $512 \times 217 \times 224$ with 20 water absorption bands, 108–112, 154–167, 224. So, a total of 224 bands were used for experiments. Fig. 4(b) and (c) shows the color composite of the Salinas image along with the corresponding ground truth class labels. Unlike the Purdue data, Salinas scene does not have an in issue in imbalanced classes as tabulated in Table II where the smallest class is class 13 with 916 data samples.

3) University of Pavia: A third and last hyperspectral image data used for experiments was the University of Pavia image, as shown in Fig. 5, which is an urban area surrounding the University of Pavia, Italy. It was recorded by the Reflective Optics System Imaging Spectrometer (ROSIS)-03 satellite sensor over an urban area surrounding the University of Pavia, Italy. It is of size $610 \times 340 \times 115$ with a spatial resolution of 1.3-m per pixel and a spectral coverage ranging from 0.43 to 0.86 μm with a spectral resolution of 4 nm (12 most noisy channels were removed before experiments). Fig. 5(b) provides its ground-truth map of nine classes along with color class labels in Fig. 5(c). Table III also tabulates the number of data samples in parentheses collected for each class. Like Salinas scene, this scene also has very large classes with only one small class less than 1000 data samples, class 9 with 947 samples. However, this scene has more complicated BKG than the other two studied scenes as already shown in [33] and [34] where the precision rate (PR) of this scene was much lower than the other two scenes.

According to the recent work [35], the Harsanyi–Farrand–Chang (HFC) method [36] along with its noise-whitened HFC method developed in [29]–[31], and [35] was used to estimate VD for the three hyperspectral images, and their estimated VD value, n_{VD}, with various false alarm probabilities, $\mathrm{P_F}$, is tabulated in Table IV.

In order to make fair comparison with the results in [37] and [38], n_{VD} was set to $n_{\mathrm{VD}} = 18$ for the Purdue data, $n_{\mathrm{VD}} = 21$ for Salinas, and $n_{\mathrm{VD}} = 14$ for the University of

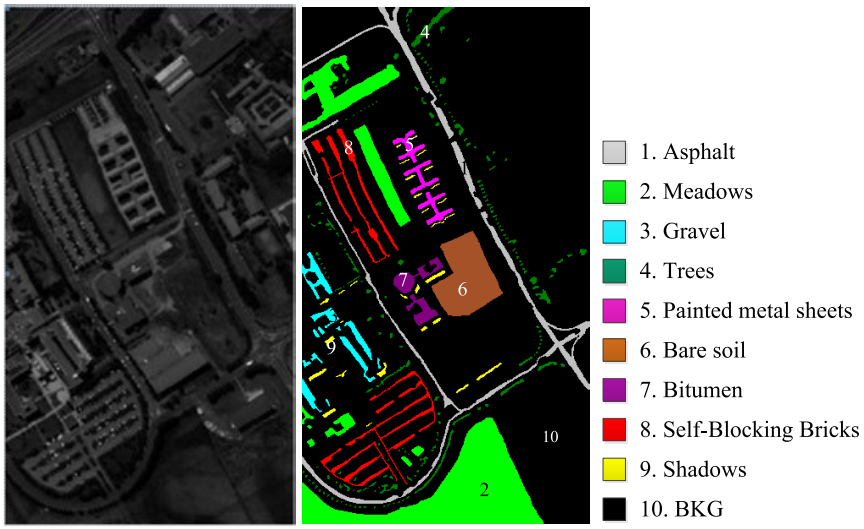
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TABLE II
CLASS LABELS OF CLASSES IN SALINAS

class 1 (2009)	Brocoli_green_weeds_1	class 10 (3278)	Corn_senesced_green_weeds
class 2 (3726)	Brocoli_green_weeds_2	class 11 (1068)	Lettuce_romaine_4wk
class 3 (1976)	Fallow	class 12 (1927)	Lettuce_romaine_5wk
class 4 (1394)	Fallow_rough_plow	class 13 (916)	Lettuce_romaine_6wk
class 5 (2678)	Fallow_smooth	class 14 (1070)	Lettuce_romaine_7wk
class 6 (3959)	Stubble	class 15 (7268)	Vinyard_untrained
class 7 (3579)	Celery	class 16 (1807)	Vinyard_vertical_trellis
class 8 (11271)	Grapes_untrained	class 17 (56975)	BKG
class 9 (6203)	Soil_vinyard_develop		

TABLE III
CLASS LABELS OF THE UNIVERSITY OF PAVIA

class 1 (6631)	Asphalt	class 5 (1345)	Painted metal sheets	Class 9 (947)	Shadows
class 2 (18649)	Meadows	class 6 (5029)	Bare Soil	Class 10 (164624)	BKG
class 3 (2099)	Gravel	class 7 (1330)	Bitumen		
class 4 (3064)	Trees	class 8 (3682)	Self-Blocking Bricks		



(a) (b) (c)

Fig. 5. University of Pavia and its ground truth with nine classes. (a) University of Pavia scene. (b) Ground-truth map. (c) Classes by colors.

Pavia in which case, n_{BS} was determined by the false alarm probability $\mathrm{P_F} = 10^{-4}$.

B. Classification Methods and Test BS Methods to Be Used for Comparison

To demonstrate the advantages of SMI-BS for HSIC, three important components are crucial to justify its utility. One is classification methods selected to perform classification. Another is test BS methods selected to conduct comparative studies and analyses. A third one is classification measures used to evaluate classification performance.

1) Classification Methods: Many HSIC methods have been reported in the literature, among which the spectral-spatial classification is one of most widely used and promising methods for HSIC. Of particular interest is an edge preserving filtering (EPF)-based approach developed in [32] which used support vector machine (SVM) as a spectral classifier followed by an EPF to perform final classification. It was later extended to iterative EPF (IEPF) methods in [39] which significantly improved EPF-based methods. Since 4 EPF methods proposed in [32] performed similarly, EPF-B-g was selected to be used for experiments. Correspondingly, IEPF-B-g was also selected for this reason. When EPF-B-g and IEPF-B-g were implemented, the training samples setup was exactly the same as those used in [32] and [39], respectively.

Another type of the spectral-spatial classification was also selected for experiments. It was developed in [40], referred to as iterative constrained energy minimization (ICEM) in [40] which used constrained energy minimization (CEM) as a subpixel classifier followed by Gaussian filters. There are three key differences between EPF-B-g and ICEM. One is their

TABLE IV
n_{VD} ESTIMATED BY HFC/NWHFC

	$\mathrm{P_F}=10^{-1}$	$\mathrm{P_F}=10^{-2}$	$\mathrm{P_F}=10^{-3}$	$\mathrm{P_F}=10^{-4}$	$\mathrm{P_F}=10^{-5}$
Purdue	73/21	49/19	35/18	27/18	25/17
Salinas	32/33	28/24	25/21	21/21	20/20
U. of Pavia	25/34	21/27	16/17	14/14	13/12

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used spectral classifiers where SVM used in EPF-B-g is a hard-decision classier compared with CEM used in ICEM which is a soft-decision classier. Another is that ICEM deploys a feedback system, while EPF-B-g does not. A third one is that EPF-B-g needs training samples which are not required by ICEM. In this case, the EPF-B-g classification results are sensitive to the selection of training samples. To avoid inconsistency in performance evaluation, EPF-B-g was performed ten runs to calculate their means to reduce uncertainty where the same set of training samples were used for each run by different BS methods for fair comparison.

2) BS Methods: Many BS methods are proposed in the past. In the following experiments, three types of BS methods are selected in comparison with SMI-BS using three BDs, SID, SAM, and SIDAM, referred to as SMI-BS-SID, SMI-BS-SAM, and SMI-BS-SIDAM, respectively. Most of BS methods to be compared have been shown to be promising and effective.

1) UBS is simple and does not need any criterion or prior knowledge. Generally, UBS is an efficient BS method.
2) BP methods
   a) Minimum estimated abundance covariance (MEAC) [13]
   b) forward class signature-constrained background suppression-BP (FCSCBS-BP) [38]
   c) backward class signature-constrained background suppression-BP (FCSCBS-BP) [38]
3) BSS methods
   a) sequential channel capacity band subset selection [6]
   b) successive channel capacity band subset selection [6]
   c) MDPP [27]
   d) DSEBS [28]
   e) linearly constrained minimum variance (LCMV)-sequential feed forward BS (LCMV-SFBS) [37]
   f) linearly constrained minimum variance-sequental backward BS (LCMV-SBBS) [37]

3) Classification Measures: The following three classification measures defined in [33] will be used for experiments. Let

M = the number of classes,

C_m = the set of data samples in the mth class by ground truth

$n_m = \sum_{j=1}^{M} n_{jm}$ = the number of data samples in C_m,

N = total number of data samples, $N = \sum_{m=1}^{M} n_m$.

n_{mm} = the number of signal samples in the mth class correctly classified into the mth class$\hat{C}_m$,

n_{jm} = the number of data samples in the mth class C_m but actually classified into the jth class, $\hat{C}_j$.

In traditional HSIC, the popular performance measurements are class accuracy (A) and overall accuracy (OA), given by

$$p_{\mathrm{A}}(C_m) = \text{accuracy of the } m \text{ th class} = \frac{n_{mm}}{\sum_{j=1}^{M} n_{jm}} = \frac{n_{mm}}{n_m} \tag{19}$$

$$\mathrm{P_{OA}} = \frac{1}{N}\sum_{m=1}^{M} n_{mm} = \sum_{m=1}^{M} \frac{n_m}{N} p_{\mathrm{A}}(C_m) \tag{20}$$

TABLE V
NUMBER OF PROMINENT PEAKS

	Purdue (n_{VD}=18)	Salinas (n_{VD}=21)	U. of Pavia (n_{VD}=14)
SID	18	11	3
SAM	15	12	3
SIDAM	18	13	2

which shows that $\mathrm{P_{OA}}$ utilizes sample ratios as weights to average the accuracy of each class. In addition, we can consider all classes to be equally likely by letting $(n_m/N) = (1/M)$ to further define $\mathrm{P_{AA}}$ as average accuracy (AA) by the number of classes, M, expressed as

$$\mathrm{P_{AA}} = \frac{1}{M}\sum_{m=1}^{M} \frac{n_{mm}}{n_m} = \frac{1}{M}\sum_{m=1}^{M} p_{\mathrm{A}}(C_m). \tag{21}$$

Another measure is PR given by

$$p_{\mathrm{PR}}(C_m) = \text{precision rate of } C_m = \frac{\hat{n}_{mm}}{\hat{n}_m} \tag{22}$$

where $\hat{n}_m = \sum_{j=1}^{M} \hat{n}_{mj}$is the total number of data samples that be classified into mth class and $\hat{n}_{mj}$ = the number of data samples classified into mth class but supposed to be in jth class. So, the average PR (APR), $\mathrm{P_{APR}}$, in correspondence to $\mathrm{P_{AA}}$ in (21) is defined by

$$\mathrm{P_{APR}} = \frac{1}{M}\sum_{m=1}^{M} \mathrm{P_{PR}}\big(\hat{C}_m\big). \tag{23}$$

Since n_{mm} is identical to $\hat{n}_{mm}$ and $\hat{N} = \sum_{i=1}^{M} \hat{n}_i$ equal to $N = \sum_{i=1}^{M} n_i$. As a result, an overall PR (OPR), $\mathrm{P_{OPR}}$, can be also defined as the counterpart of $\mathrm{P_{OA}}$ in (20) by

$$\begin{aligned}\mathrm{P_{OPR}} &= \sum_{i=1}^{M} p\big(\hat{C}_i\big)\mathrm{P_{PR}}\big(\hat{C}_i\big) = \sum_{i=1}^{M}\left(\frac{\hat{n}_i}{\hat{N}}\right)\mathrm{P_{PR}}\big(\hat{C}_i\big) \\ &= \sum_{i=1}^{M}\left(\frac{n_i}{N}\right)\mathrm{P_A}(C_i) = \mathrm{P_{OA}}\end{aligned} \tag{24}$$

which implies that $\mathrm{P_{OPR}}$ is essentially the same as $\mathrm{P_{OA}}$.

It should be noted that PR, also known as user's accuracy, is proposed to address background (BKG) issue, while many existing classification methods have removed all unlabeled data samples including BKG samples from classification. PR is the one that can include all data samples for evaluation.

VIII. Experimental Results and Discussion

Figs. 6–8 plot SMI curves produced by SMI-BS-SID, SMI-BS-SAM, and SMI-BS-SIDAM for the Purdue data, Salinas, and the University of Pavia, respectively, where prominent peaks are marked by red dots. Each corresponding number of prominent peaks, n_{PK}, was also found using the algorithm described in Section V with the threshold set to 10^{-2} and tabulated in Table V.

It should be noted that while determining n_{PK} we can also find the bands corresponding to each of prominent peaks marked by red dots.

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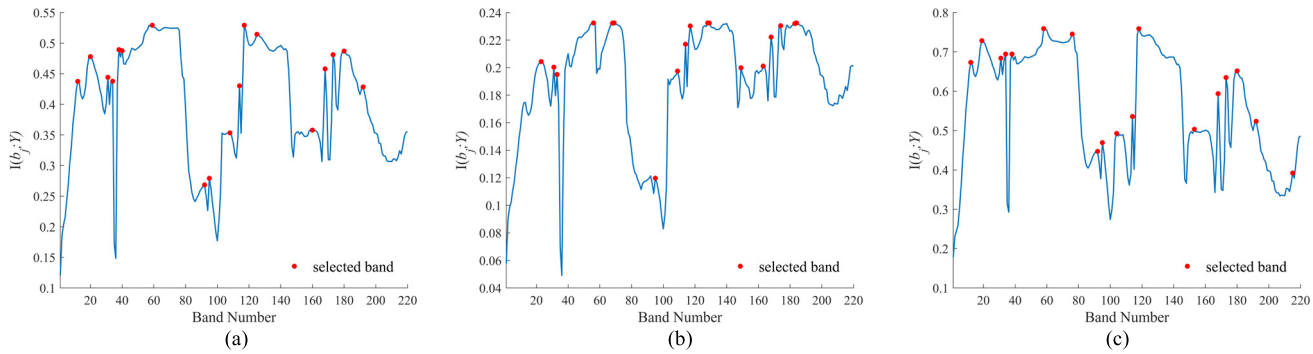


Fig. 6. SMI curves produced by SMI using SID, SAM, and SIDAM with 18 selected bands for the Purdue data. (a) SID. (b) SAM. (c) SIDAM.

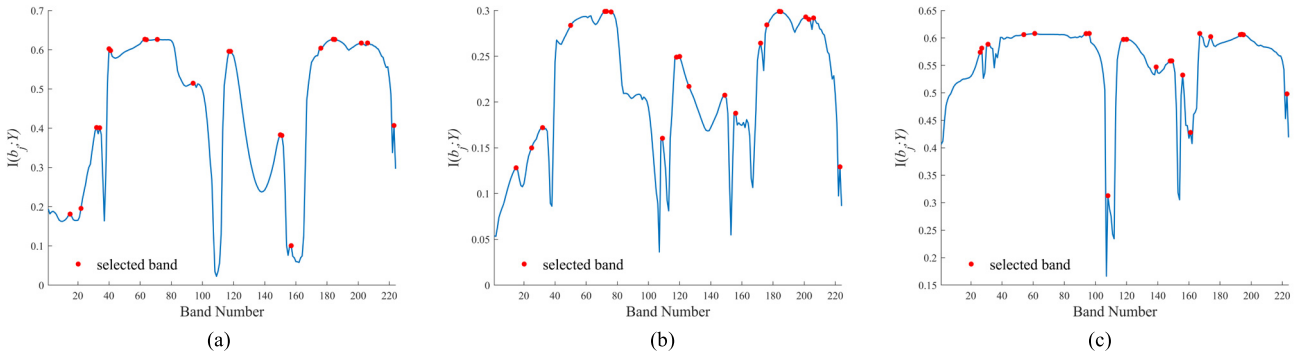


Fig. 7. SMI curves yield by SID, SAM, and SIDAM and the 21 selected bands for Salinas. (a) SID. (b) SAM. (c) SIDAM.

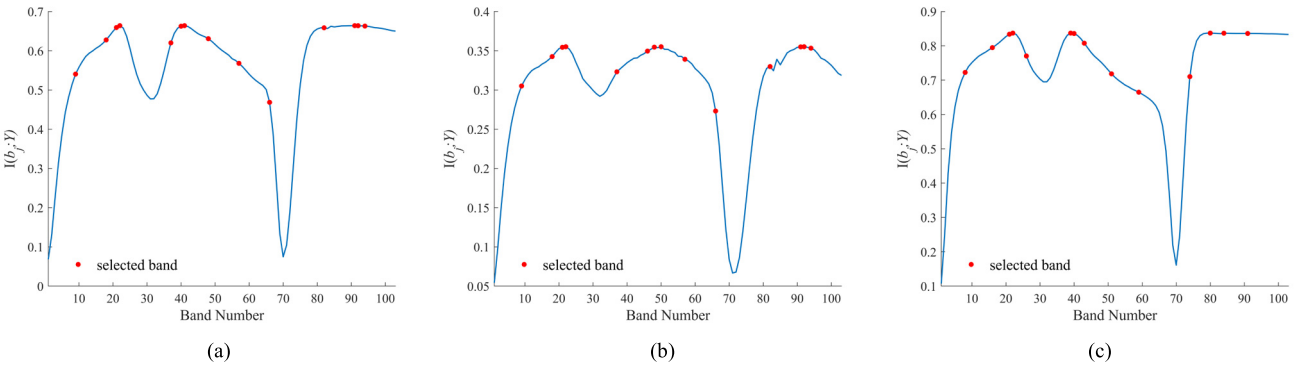


Fig. 8. SMI curves yield by SID, SAM, and SIDAM and the 14 selected bands for the University of Pavia. (a) SID. (b) SAM. (c) SIDAM.

According to Table V and Figs. 6–8, the SMI curves of the three data sets behaved quite different. In particular, the numbers of prominent peaks found by three BD measures for Salinas and the University of Pavia were less than their corresponding n_{VD}. More interestingly, the University of Pavia had only two prominent peaks shown in Fig. 8, where its SMI curve was much smoother than the SMI curves found for the Purdue data and Salinas. As discussed in Section V, a prominent peak not only represents its corresponding band with a higher BD but also indicates its band correlation with other bands is low. Due to the fact that the University of Pavia image was obtained by ROSIS sensor with spectral resolution (4 nm) which is much higher than the other two images (10 nm), the University of Pavia has also much higher correlation among bands than that for the other two images. This may explain why the University of Pavia has smoother SMI curves with only two peaks compared with the SMI curves obtained for the Purdue data and Salinas.

From the red dots in Figs. 6–8 which specify the PBs, Tables VI–VIII list bands selected by SMI-BS (SIDAM, SID, SAM) along with UBS, SQ-CCBSS [6], SC-CCBSS [6], MEAC [32], MDPP [27], DSEBS [28], LCMV-SFBS [38], LCMV-SBBS [38], FCSCBS-BP [37], and BCSCBS-BP [37]. From these tables, the bands selected by SMI-BS were much dispersive than the bands selected by other methods such as SQ-CCBSS, SC-CCBSS, and FCSCBS-BP.

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TABLE VI
BAND SELECTED BY VARIOUS BS METHODS FOR THE PURDUE DATA

	Selected bands (n_{VD} = 18)																	
SMI-BS (SIDAM)	58	118	76	19	38	34	31	12	180	173	168	114	192	153	104	95	92	215
SMI-BS (SID)	59	117	125	38	40	180	173	20	168	31	34	12	114	192	160	108	95	92
SMI-BS (SAM)	23	31	33	56	69	95	109	114	117	129	149	163	168	174	184	68	128	183
SQ-CCBSS	12	44	48	47	53	43	16	14	46	52	49	13	50	45	51	17	15	11
SC-CCBSS	195	196	175	190	41	48	44	42	43	45	193	47	200	176	197	199	198	194
MEAC	159	92	3	96	82	36	39	55	41	1	2	33	206	38	163	17	204	9
MDPP	10	39	59	75	79	85	92	130	140	146	147	149	150	152	160	164	175	193
DSEBS	42	129	97	131	174	16	176	177	172	43	192	193	98	171	99	132	40	33
LCMV-SFBS	29	42	64	35	37	32	34	104	107	158	80	151	152	103	161	220	65	106
LCMV-SBBS	7	86	46	3	156	150	42	154	152	108	151	157	85	84	160	32	159	219
FCSCBS-BP	29	28	30	32	27	26	25	24	33	31	23	22	21	20	19	34	18	120
BCSCBS-BP	5	35	29	11	3	42	12	96	121	4	101	158	65	31	213	99	165	26
UBS	1	14	27	40	53	66	79	92	105	118	131	144	157	170	183	196	209	220

TABLE VII
BAND SELECTED BY VARIOUS BS METHODS FOR SALINAS

	Selected bands (n_{VD} = 21)																				
SMI-BS (SIDAM)	27	31	96	108	120	139	149	156	161	167	174	194	223	26	54	61	94	118	148	193	195
SMI-BS (SID)	15	32	34	40	63	117	150	157	184	206	223	22	41	64	71	94	118	151	176	185	202
SMI-BS (SAM)	15	32	73	109	120	149	156	172	184	201	206	223	25	50	72	76	118	126	176	185	203
SQ-CCBSS	50	48	185	182	51	206	183	47	42	49	44	184	52	179	43	178	46	186	181	180	45
SC-CCBSS	156	128	155	119	194	118	117	192	191	193	122	123	195	196	197	121	124	120	125	198	116
MEAC	107	148	203	149	5	8	105	3	28	12	18	10	44	36	25	17	51	32	110	68	58
MDPP	1	8	11	22	27	28	50	57	58	65	90	99	105	119	123	134	142	157	175	191	204
DSEBS	99	101	16	119	177	112	44	46	120	47	131	175	196	121	17	102	174	180	187	135	42
LCMV-SFBS	44	32	160	38	56	157	156	28	158	47	159	155	3	14	64	107	108	46	109	163	96
LCMV-SBBS	2	20	19	21	18	17	22	23	41	24	25	16	26	39	38	10	42	3	40	9	14
FCSCBS-BP	44	45	43	46	47	48	42	49	50	51	52	53	54	55	56	57	58	59	41	60	61
BCSCBS-BP	175	120	173	174	106	176	121	32	197	220	84	38	44	196	105	192	12	195	10	117	156
UBS	1	12	23	34	45	56	67	78	89	100	111	122	133	144	155	166	177	188	199	210	224

TABLE VIII
BAND SELECTED BY VARIOUS BS METHODS FOR THE UNIVERSITY OF PAVIA

	Selected bands (n_{VD} = 14)													
SMI-BS (SIDAM)	22	39	8	16	21	26	40	43	51	59	74	80	84	91
SMI-BS (SID)	22	41	91	9	18	21	37	40	48	57	66	82	92	94
SMI-BS (SAM)	22	50	92	9	18	21	37	46	48	57	66	82	91	94
SQ-CCBSS	16	81	89	80	19	18	17	84	20	79	88	83	87	82
SC-CCBSS	88	91	89	95	87	56	57	58	55	54	92	93	94	90
MEAC	1	23	24	40	42	58	56	59	48	31	47	83	25	54
MDPP	2	23	44	46	50	62	66	73	89	91	92	93	96	102
DSEBS	86	102	64	20	1	63	65	6	19	22	7	66	95	67
LCMV-SFBS	16	91	62	74	1	57	43	86	55	56	92	63	4	36
LCMV-SBBS	1	2	3	85	83	86	82	12	73	72	87	6	11	40
FCSCBS-BP	16	15	14	17	18	13	19	12	20	11	21	10	22	9
BCSCBS-BP	84	83	66	67	77	82	1	81	78	79	68	76	101	69
UBS	1	9	17	25	33	41	49	57	65	73	81	89	97	103

This is because the use of prominent peaks to guarantee the interband correlation among selected bands was low.

A. Purdue Indiana Indian Pines

Tables IX–XI show the results of P_{AA}, P_{OA}, and P_{APR} produced by EPF-B-g, IEPF-B-g, and ICEM using the bands selected in Table VI for the Purdue data where the best results are boldfaced by red. In addition, those results which were better than that produced by using full bands are also boldfaced by black for comparison. From Table IX, when EPF-B-g was used for classification, the values of P_{AA}, $P_{OA,}$ and P_{APR} using SMI-BS-SIDAM and SMI-BS-SID were higher than that using full bands and SMI-BS-SID was the best. Specifically, EPF-B-g was significantly improved by IEPF-B-g in Table X. Also, the classification accuracies were significantly increased using most of BS methods compared with using full bands except three cases which were only slightly worse in P_{OA} and P_{AA} but still better in P_{APR} in Table X. Interestingly, when ICEM was used for classification, except LCMV-SBBS, all other BS methods could not improve their results over using full bands in Table XI. An interesting observation

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TABLE IX

P_{OA}, P_{APR}, AND P_{AA} PRODUCED BY EPF-B-g USING BANDS SELECTED IN TABLE VI FOR THE PURDUE DATA

class	SMI-BS-SIDAM	SMI-BS-SID	SMI-BS-SAM	SQ-CCBSS	SC-CCBSS	MEAC	MDPP	DSEBS	LCMV-SFBS	LCMV-SBBS	FCSCBS-BP	BCSCBS-BP	UBS	full bands
1	0.9935	0.9739	0.9870	0.9978	0.9760	1.0000	1.0000	0.9804	0.9870	0.8609	0.9652	0.9913	0.9978	0.9978
2	0.8641	0.8839	0.8291	0.4364	0.5963	0.8001	0.7491	0.8416	0.6062	0.5014	0.7954	0.8076	0.8047	0.8222
3	0.8304	0.8406	0.7466	0.6271	0.6686	0.7948	0.6477	0.8482	0.7188	0.6157	0.5219	0.7577	0.7393	0.7959
4	0.9920	0.9958	0.9886	0.9991	0.9928	0.9954	0.9962	0.9949	0.9679	0.9966	0.9156	0.9945	0.9949	0.9932
5	0.9507	0.9487	0.9501	0.9099	0.9318	0.9290	0.9325	0.9393	0.9253	0.8455	0.9383	0.9381	0.9335	0.9436
6	0.9984	0.9986	0.9967	0.9846	0.9873	0.9996	0.9900	0.9901	0.9993	0.9781	0.9990	0.9989	0.9936	0.9982
7	0.9643	0.9536	0.9429	0.9071	0.3571	0.9679	0.8429	0.9286	0.8571	0.6500	0.9571	0.9286	0.9500	0.9321
8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
9	0.6650	0.8400	0.6600	0.3500	0.4850	0.7900	0.8500	0.8800	0.3700	0.1650	0.5850	0.9250	0.8850	0.7300
10	0.8824	0.9113	0.8867	0.7682	0.6371	0.8063	0.7485	0.8883	0.6767	0.6808	0.8336	0.8470	0.8160	0.9060
11	0.8922	0.8970	0.8475	0.7519	0.8286	0.8912	0.8523	0.8998	0.7692	0.7611	0.7535	0.8548	0.8587	0.9057
12	0.9715	0.9782	0.9184	0.6563	0.8915	0.8774	0.9501	0.9747	0.6415	0.3642	0.7165	0.9388	0.9639	0.9424
13	0.9922	0.9937	0.9937	0.9985	0.9931	0.9956	0.9951	0.9951	0.9951	0.9980	0.9956	0.9946	0.9946	0.9931
14	0.9641	0.9680	0.9743	0.9624	0.9661	0.9823	0.9632	0.9887	0.9640	0.9849	0.9732	0.9817	0.9640	0.9622
15	0.9611	0.9425	0.8723	0.6147	0.5647	0.9360	0.9575	0.8984	0.8736	0.7016	0.9484	0.8961	0.9831	0.9886
16	0.9892	0.9849	0.9978	0.9956	0.9967	0.9978	0.9925	0.9925	0.9935	0.9935	1	0.9989	0.9967	0.9989
P_{OA}	**0.9192**	**0.9270**	0.8918	0.7639	0.8073	0.8953	0.8634	0.9200	0.8003	0.7498	0.8302	0.8910	0.8890	0.9150
P_{APR}	**0.9338**	**0.9366**	0.9102	0.8118	0.8396	0.9168	0.9003	0.9352	0.8603	0.7780	0.8688	0.9123	0.9191	0.9276
P_{AA}	**0.9319**	**0.9444**	0.9120	0.8100	0.8046	0.9227	0.9042	0.9400	0.8341	0.7561	0.8687	0.9284	0.9298	0.9319

TABLE X

P_{OA}, P_{APR}, AND P_{AA} PRODUCED BY IEPF-B-g USING BANDS SELECTED IN TABLE VI FOR THE PURDUE DATA

class	SMI-BS-SIDAM	SMI-BS-SID	SMI-BS-SAM	SQ-CCBSS	SC-CCBSS	MEAC	MDPP	DSEBS	LCMV-SFBS	LCMV-SBBS	FCSCBS-BP	BCSCBS-BP	UBS	full bands
1	1.0000	1.0000	1.0000	0.9957	0.9957	1.0000	1.0000	0.9957	1.0000	1.0000	0.9913	1.0000	1.0000	0.9957
2	0.9678	0.9655	0.9702	0.9744	0.9789	0.9694	0.9730	0.9813	0.9452	0.9279	0.9641	0.9617	0.9512	0.9478
3	0.9876	0.9894	0.9890	0.9753	0.9888	0.9892	0.9900	0.9786	0.9816	0.9786	0.9698	0.9849	0.9892	0.9816
4	0.9962	0.9954	0.9962	0.9945	0.9966	0.9992	0.9975	0.9928	0.9932	0.9992	0.9848	0.9966	0.9983	0.9962
5	0.9805	0.9795	0.9787	0.9679	0.9756	0.9783	0.9787	0.9735	0.9766	0.9716	0.9679	0.9747	0.9766	0.9752
6	0.9996	0.9996	0.9995	0.9818	0.9952	0.9999	0.9989	0.9923	0.9992	0.9970	0.9959	0.9995	0.9989	0.9982
7	1.0000	0.9929	0.9929	0.9821	0.9893	0.9929	0.9857	0.9857	0.9964	0.9893	0.9929	1.0000	0.9929	0.9893
8	1.0000	1.0000	1.0000	0.9983	0.9994	1.0000	1.0000	0.9990	1.0000	1.0000	0.9987	1.0000	1.0000	1.0000
9	1.0000	1.0000	1.0000	0.9950	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9850	1.0000	1.0000	1.0000
10	0.9740	0.9754	0.9728	0.9568	0.9782	0.9689	0.9641	0.9733	0.9336	0.9362	0.9668	0.9787	0.9687	0.9612
11	0.9647	0.9680	0.9629	0.9690	0.9716	0.9647	0.9704	0.9624	0.9304	0.9411	0.9411	0.9595	0.9609	0.9413
12	0.9887	0.9907	0.9879	0.9889	0.9845	0.9867	0.9901	0.9843	0.9853	0.9747	0.9752	0.9860	0.9892	0.9912
13	0.9971	0.9966	0.9971	0.9971	0.9932	0.9976	0.9971	0.9888	0.9976	0.9980	0.9917	0.9971	0.9971	0.9961
14	0.9988	0.9992	0.9994	0.9749	0.9945	0.9983	0.9989	0.9924	0.9959	0.9955	0.9916	0.9983	0.9990	0.9943
15	0.9977	0.9972	0.9982	0.9697	0.9951	0.9951	0.9984	0.9858	0.9940	0.9956	0.9889	0.9953	0.9984	0.9946
16	0.9978	0.9978	0.9989	0.9849	0.9935	0.9957	0.9989	0.9925	0.9989	0.9957	0.9935	0.9968	1.0000	0.9957
P_{OA}	**0.9816**	**0.9824**	**0.9814**	**0.9747**	**0.9837**	**0.9811**	**0.9829**	**0.9794**	0.9649	0.9642	0.9699	**0.9791**	**0.9779**	0.9705
P_{APR}	**0.9810**	**0.9761**	**0.9709**	**0.9847**	**0.9809**	**0.9812**	**0.9793**	**0.9856**	0.9729	0.9695	0.9749	**0.9691**	**0.9762**	0.9651
P_{AA}	**0.9907**	**0.9904**	**0.9902**	**0.9816**	**0.9894**	**0.9897**	**0.9901**	**0.9861**	0.9830	0.9813	0.9812	**0.9893**	**0.9888**	0.9849

TABLE XI

P_{OA}, P_{APR}, AND P_{AA} PRODUCED BY ICEM USING BANDS SELECTED IN TABLE VI FOR THE PURDUE DATA

class	SMI-BS-SIDAM	SMI-BS-SID	SMI-BS-SAM	SQ-CCBSS	SC-CCBSS	MEAC	MDPP	DSEBS	LCMV-SFBS	LCMV-SBBS	FCSCBS-BP	BCSCBS-BP	UBS	full bands
1	0.9782	1.0000	0.9783	0.9782	0.9565	0.9565	0.9565	0.9782	0.9782	0.9782	0.9783	0.9565	0.9565	1
2	0.9684	0.9754	0.9762	0.9586	0.9740	0.9642	0.9747	0.9789	0.9663	0.9705	0.9594	0.9692	0.9740	0.9733
3	0.9698	0.9783	0.9855	0.9686	0.9783	0.9722	0.9831	0.9795	0.9710	0.9771	0.9843	0.9819	0.9807	0.9807
4	0.9915	0.9873	0.9873	0.9915	0.9915	0.9789	0.9831	0.9915	0.9831	0.9957	0.9916	0.9831	0.9873	0.9873
5	0.9420	0.9544	0.9648	0.9358	0.9689	0.9296	0.9710	0.9710	0.9648	0.9751	0.9627	0.9482	0.9565	0.9585
6	0.9643	0.9739	0.9767	0.9684	0.9520	0.9849	0.9821	0.9712	0.9726	0.9643	0.9603	0.9767	0.9698	0.9739
7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
8	0.9958	0.9854	0.9958	0.9958	0.9937	0.9895	0.9979	0.9979	0.9979	0.9979	0.9979	0.9937	0.9916	0.9958
9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
10	0.9362	0.9259	0.9403	0.9495	0.9650	0.9403	0.9495	0.9629	0.9701	0.9681	0.9516	0.9743	0.9259	0.9578
11	0.9723	0.9809	0.9735	0.9714	0.9706	0.9625	0.9808	0.9670	0.9702	0.9816	0.9666	0.9752	0.9792	0.9804
12	0.9679	0.9663	0.9730	0.9780	0.9763	0.9814	0.9645	0.9814	0.9679	0.9629	0.9781	0.9798	0.9763	0.9780
13	0.9804	0.9756	0.9854	0.9658	0.9853	0.9853	0.9902	0.9853	0.9804	0.9902	0.9756	0.9854	0.9756	0.9853
14	0.9509	0.9375	0.9360	0.9533	0.9343	0.9525	0.9565	0.9478	0.9549	0.9707	0.9597	0.9613	0.9573	0.9620
15	0.9352	0.9482	0.9585	0.9715	0.9689	0.9326	0.9611	0.9585	0.9766	0.9844	0.9663	0.9689	0.9740	0.9818
16	0.9784	0.9892	0.9892	0.9892	0.9892	0.9784	0.9784	0.9892	0.9784	0.9892	0.9892	0.9785	0.9892	0.9892
P_{OA}	0.9638	0.9660	0.9682	0.9655	0.9677	0.9626	0.9732	0.9707	0.9699	**0.9760**	0.9674	0.9733	0.9696	0.9747
P_{APR}	0.8748	0.8850	0.8881	0.8814	0.9034	0.8857	0.8913	0.8940	0.8912	**0.8996**	0.8908	0.8859	0.8848	0.8926
P_{AA}	0.9708	0.9737	0.9763	0.9735	0.9753	0.9693	0.9769	0.9788	0.9771	**0.9817**	0.9764	0.9770	0.9747	0.9815

from Table XI is that ICEM did very well and better than EPF-B-g and IEPF-B-g for classification of small classes, such as classes 7 and 9 with 100% accuracy of P_A across board. This made sense since ICEM is a mixed pixel spectral classification and was able to capture spectral information of data samples in small classes which is crucial.

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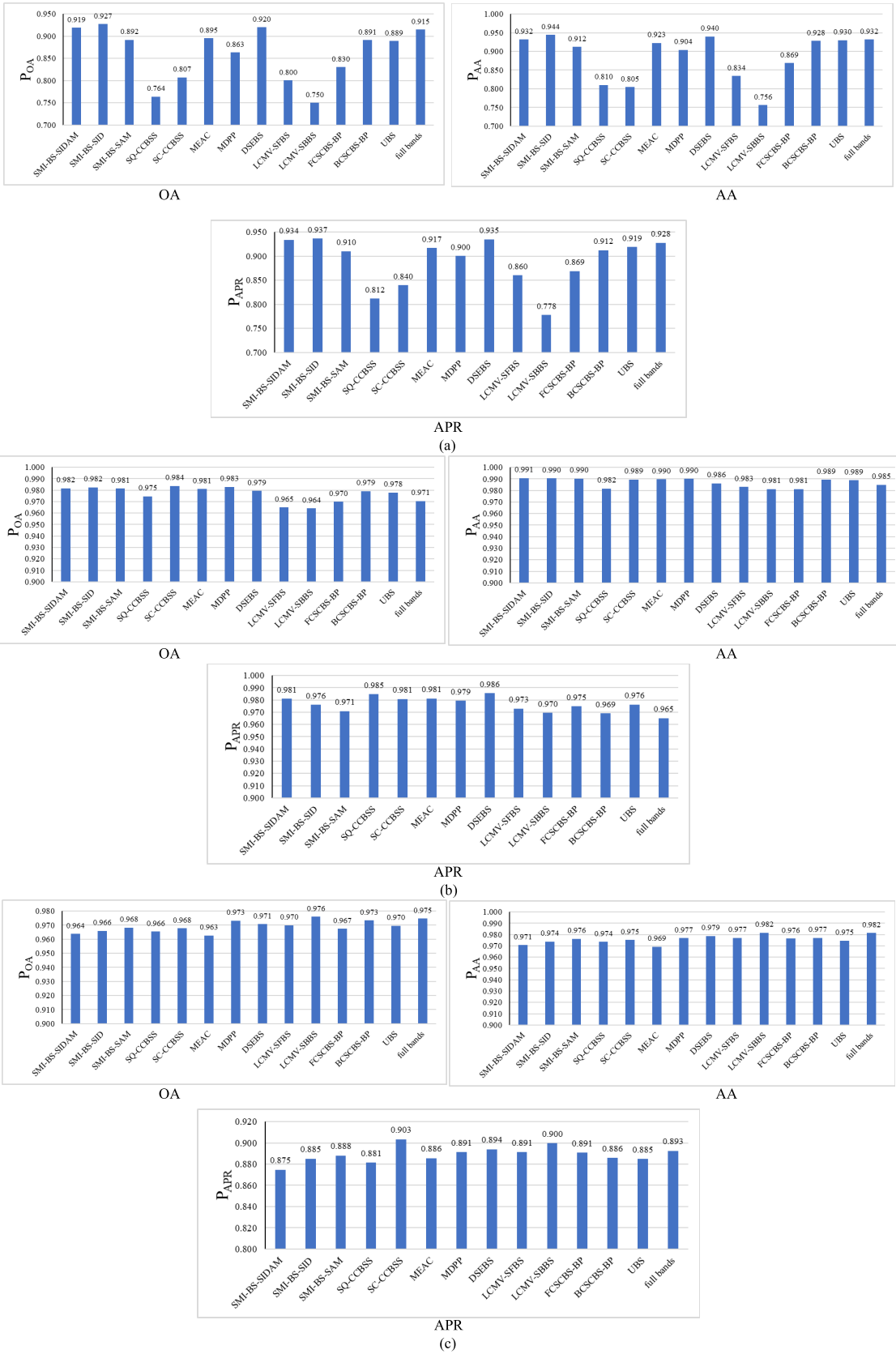


Fig. 9. Histograms of P_{OA}, P_{AA}, and P_{APR} produced by EPF-B-g, IEPF-B-g, and ICEM for the Purdue data. (a) EPF-B-g. (b) IEPF-B-g. (c) ICEM.

To better present the comparative results in Tables IX–XI for visual display, Fig. 9 plots the histograms of P_{OA}, P_{AA}, and P_{APR} produced by the three classifiers, EPF-B-g, IEPF-B-g, and ICEM in Tables IX–XI for the Purdue data where the SMI-BS methods performed consistently well for all the three classifiers compared with other BS methods.

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TABLE XII

P_{OA}, P_{APR}, AND P_{AA} PRODUCED BY EPF-B-g USING BANDS SELECTED IN TABLE VII FOR SALINAS

class	SMI-BS-SIDAM	SMI-BS-SID	SMI-BS-SAM	SQ-CCBSS	SC-CCBSS	MEAC	MDPP	DSEBS	LCMV-SFBS	LCMV-SBBS	FCSCBS-BP	BCSCBS-BP	UBS	full bands
1	1.0000	0.9999	0.9998	1.0000	1.0000	0.9984	1.0000	1.0000	0.9992	0.9989	0.9947	0.9997	1.0000	1.0000
2	0.9999	1.0000	1.0000	0.9987	1.0000	1.0000	1.0000	0.9999	1.0000	1.0000	0.9930	1.0000	1.0000	1.0000
3	0.9979	0.9997	0.9982	0.9957	0.9949	0.9995	1.0000	1.0000	0.9512	0.9793	0.9942	1.0000	0.9984	0.9999
4	0.9994	0.9998	0.9992	0.9996	0.9987	0.9996	0.9999	0.9997	0.9985	0.9994	1.0000	0.9996	0.9996	0.9996
5	0.9856	0.9848	0.9844	0.9897	0.9820	0.9825	0.9858	0.9835	0.9441	0.9581	0.9744	0.9854	0.9831	0.9830
6	1.0000	1.0000	1.0000	0.9998	0.9999	1.0000	0.9999	1.0000	1.0000	1.0000	0.9991	1.0000	1.0000	1.0000
7	1.0000	0.9996	1.0000	0.9968	0.9887	1.0000	0.9996	1.0000	1.0000	0.9993	0.9948	0.9999	1.0000	0.9999
8	0.8338	0.8385	0.8472	0.7444	0.8381	0.8514	0.8691	0.8860	0.8098	0.8092	0.8438	0.8698	0.8443	0.8286
9	0.9995	0.9995	0.9996	0.9974	0.9978	0.9938	0.9998	0.9997	0.9544	0.9738	0.9589	0.9995	0.9993	0.9979
10	0.9563	0.9681	0.9635	0.9394	0.9372	0.9632	0.9692	0.9753	0.9213	0.9640	0.9282	0.9641	0.9662	0.9660
11	0.9949	0.9966	0.9978	0.9889	0.9906	0.9921	0.9962	0.9976	0.9730	0.9940	0.9779	0.9977	0.9944	0.9971
12	1.0000	1.0000	1.0000	1.0000	0.9998	1.0000	1.0000	1.0000	1.0000	1.0000	0.9906	1.0000	1.0000	1.0000
13	0.9934	0.9905	0.9915	0.9849	0.9867	0.9903	0.9956	0.9934	0.9883	0.9910	0.9807	0.9962	0.9932	0.9901
14	0.9921	0.9956	0.9948	0.9807	0.9953	0.9955	0.9922	0.9965	0.9954	0.9957	0.9916	0.9970	0.9943	0.9940
15	0.7819	0.8167	0.7702	0.7411	0.7547	0.8339	0.7540	0.7824	0.7348	0.8230	0.7352	0.7779	0.7784	0.8499
16	0.9971	0.9965	0.9965	0.9897	0.9972	0.9921	0.9919	0.9968	0.9911	0.9923	0.9811	0.9970	0.9958	0.9951
P_{OA}	0.9321	0.9385	0.9338	0.9059	0.9268	**0.9422**	0.9365	0.9443	0.9091	0.9278	0.9184	0.9398	0.9343	0.9405
P_{APR}	0.9667	0.9705	0.9684	0.9542	0.9617	**0.9691**	0.9704	0.9732	0.9393	0.9564	0.9478	0.9721	0.9685	0.9689
P_{AA}	0.9707	0.9741	0.9714	0.9592	0.9664	**0.9745**	0.9721	0.9757	0.9538	0.9674	0.9586	0.9740	0.9717	0.9751

TABLE XIII

P_{OA}, P_{APR}, AND P_{AA} PRODUCED BY IEPF-B-g USING BANDS SELECTED IN TABLE VII FOR SALINAS

class	SMI-BS-SIDAM	SMI-BS-SID	SMI-BS-SAM	SQ-CCBSS	SC-CCBSS	MEAC	MDPP	DSEBS	LCMV-SFBS	LCMV-SBBS	FCSCBS-BP	BCSCBS-BP	UBS	full bands
1	0.9999	1.0000	1.0000	0.9999	1.0000	0.9992	1.0000	1.0000	1.0000	1.0000	0.9988	0.9966	0.9999	1.0000
2	0.9970	0.9963	0.9981	0.9954	0.9999	0.9966	0.9977	0.9972	0.9969	0.9971	0.9953	0.9976	0.9972	0.9969
3	0.9992	0.9997	0.9998	0.9977	0.9987	0.9992	0.9998	0.9990	0.9996	0.9998	0.9973	0.9963	0.9999	0.9990
4	0.9984	0.9984	0.9978	0.9986	0.9986	0.9978	0.9982	0.9973	0.9981	0.9971	0.9973	0.9966	0.9987	0.9974
5	0.9931	0.9920	0.9924	0.9922	0.9926	0.9918	0.9925	0.9917	0.9879	0.9822	0.9821	0.9890	0.9925	0.9903
6	0.9989	0.9984	0.9991	0.9984	0.9988	0.9989	0.9993	0.9988	0.9993	0.9990	0.9972	0.9953	0.9991	0.9979
7	0.9957	0.9957	0.9972	0.9946	0.9965	0.9948	0.9970	0.9962	0.9958	0.9954	0.9944	0.9953	0.9967	0.9960
8	0.8927	0.8788	0.8890	0.8359	0.8643	0.9094	0.8954	0.9177	0.8668	0.8641	0.8594	0.9000	0.8880	0.8977
9	0.9965	0.9959	0.9975	0.9943	0.9956	0.9985	0.9984	0.9982	0.9993	0.9986	0.9971	0.9947	0.9976	0.9979
10	0.9911	0.9894	0.9889	0.9852	0.9818	0.9918	0.9908	0.9942	0.9923	0.9924	0.9792	0.9876	0.9936	0.9859
11	0.9977	0.9965	0.9980	0.9981	0.9976	0.9976	0.9988	0.9985	0.9991	0.9989	0.9963	0.9957	0.9987	0.9970
12	0.9999	1.0000	1.0000	0.9996	1.0000	0.9999	1.0000	1.0000	1.0000	1.0000	0.9979	0.9995	1.0000	1.0000
13	0.9944	0.9949	0.9948	0.9928	0.9992	0.9933	0.9957	0.9952	0.9945	0.9947	0.9910	0.9981	0.9945	0.9962
14	0.9939	0.9939	0.9927	0.9912	0.9950	0.9931	0.9949	0.9946	0.9946	0.9936	0.9934	0.9932	0.9941	0.9936
15	0.9335	0.9020	0.8974	0.8758	0.8670	0.9072	0.8942	0.9178	0.8858	0.8841	0.9047	0.9130	0.8738	0.9143
16	0.9899	0.9921	0.9919	0.9901	0.9894	0.9913	0.9919	0.9916	0.9975	0.9928	0.9868	0.9868	0.9910	0.9907
P_{OA}	**0.9662**	0.9588	0.9608	0.9456	0.9510	**0.9662**	0.9620	0.9697	0.9549	0.9535	0.9535	0.9637	0.9576	0.9643
P_{APR}	**0.9822**	0.9791	0.9796	0.9733	0.9763	**0.9815**	0.9812	0.9840	0.9782	0.9759	0.9763	0.9823	0.9792	0.9797
P_{AA}	**0.9857**	0.9827	0.9834	0.9775	0.9797	**0.9850**	0.9840	0.9867	0.9817	0.9806	0.9793	0.9834	0.9822	0.9844

TABLE XIV

P_{OA}, P_{APR}, AND P_{AA} PRODUCED BY ICEM USING BANDS SELECTED IN TABLE VII FOR SALINAS

class	SMI-BS (SIDAM)	SMI-BS (SID)	SMI-BS (SAM)	SQ-CCBSS	SC-CCBSS	MEAC	MDPP	DSEBS	LCMV-SFBS	LCMV-SBBS	FCSCBS-BP	BCSCBS-BP	UBS	full bands
1	0.9776	0.9781	0.9841	0.9741	0.9780	0.9761	0.9816	0.9771	0.9781	0.9746	0.9766	0.9726	0.9766	0.9751
2	0.9812	0.9826	0.9863	0.9849	0.9814	0.9873	0.9860	0.9839	0.9879	0.9855	0.9852	0.9858	0.9844	0.9868
3	0.9554	0.9352	0.9413	0.9195	0.9342	0.9175	0.9509	0.9332	0.9266	0.8907	0.8497	0.9261	0.9286	0.9459
4	0.9361	0.9491	0.9577	0.9461	0.9282	0.9383	0.9433	0.9484	0.9017	0.9139	0.8938	0.9412	0.9347	0.9620
5	0.9507	0.9459	0.9619	0.9492	0.9551	0.9622	0.9556	0.9619	0.9544	0.9559	0.9037	0.9544	0.9541	0.9604
6	0.9785	0.9816	0.9808	0.9717	0.9810	0.9780	0.9790	0.9773	0.9806	0.9884	0.9760	0.9740	0.9755	0.9775
7	0.9818	0.9880	0.9849	0.9770	0.9608	0.9854	0.9818	0.9849	0.9860	0.9790	0.9832	0.9796	0.9855	0.9855
8	0.9658	0.9642	0.9603	0.9567	0.9644	0.9599	0.9501	0.9569	0.9660	0.9641	0.9785	0.9608	0.9636	0.9722
9	0.9571	0.9581	0.9531	0.9592	0.9492	0.9508	0.9453	0.9499	0.9532	0.9231	0.9616	0.9449	0.9383	0.9723
10	0.9664	0.9610	0.9610	0.9554	0.9575	0.9472	0.9576	0.9576	0.9542	0.9518	0.9497	0.9567	0.9606	0.9683
11	0.9728	0.9813	0.9757	0.9756	0.9897	0.9363	0.9766	0.9700	0.9401	0.9157	0.9522	0.9728	0.9794	0.9775
12	0.9351	0.9600	0.9689	0.9423	0.9621	0.9693	0.9621	0.9709	0.9725	0.9735	0.9206	0.9730	0.9735	0.9839
13	0.9606	0.9793	0.9618	0.9650	0.9672	0.9661	0.9574	0.9672	0.9672	0.9629	0.9705	0.9683	0.9607	0.9716
14	0.9813	0.9794	0.9785	0.9831	0.9869	0.9747	0.9785	0.9757	0.9860	0.9710	0.9935	0.9692	0.9738	0.9888
15	0.9576	0.9490	0.9565	0.9482	0.9464	0.9645	0.9432	0.9469	0.9622	0.9666	0.9670	0.9501	0.9448	0.9677
16	0.9867	0.9779	0.9900	0.9883	0.9701	0.9778	0.9856	0.9889	0.9867	0.9851	0.9851	0.9873	0.9862	0.9845
P_{OA}	0.9653	0.9647	0.9661	0.9604	0.9611	0.9631	0.9599	0.9623	0.9648	0.9594	0.9610	0.9614	0.9613	0.9732
P_{APR}	0.9425	0.9359	0.9391	0.9352	0.9391	0.9328	0.9352	0.9363	0.9385	0.9334	0.9295	0.9339	0.9379	0.9445
P_{AA}	0.9653	0.9669	0.9689	0.9623	0.9633	0.9620	0.9647	0.9657	0.9627	0.9564	0.9529	0.9635	0.9638	0.9737

An interesting observation from Fig. 9 is that when a classifier is less effective such as EFP-B-g, BS methods generally show advantages over using full bands. On the other hand, if a classifier is very effective such as IEPF-B-g and ICEM, BS methods become losing their competitivity to using full bands.

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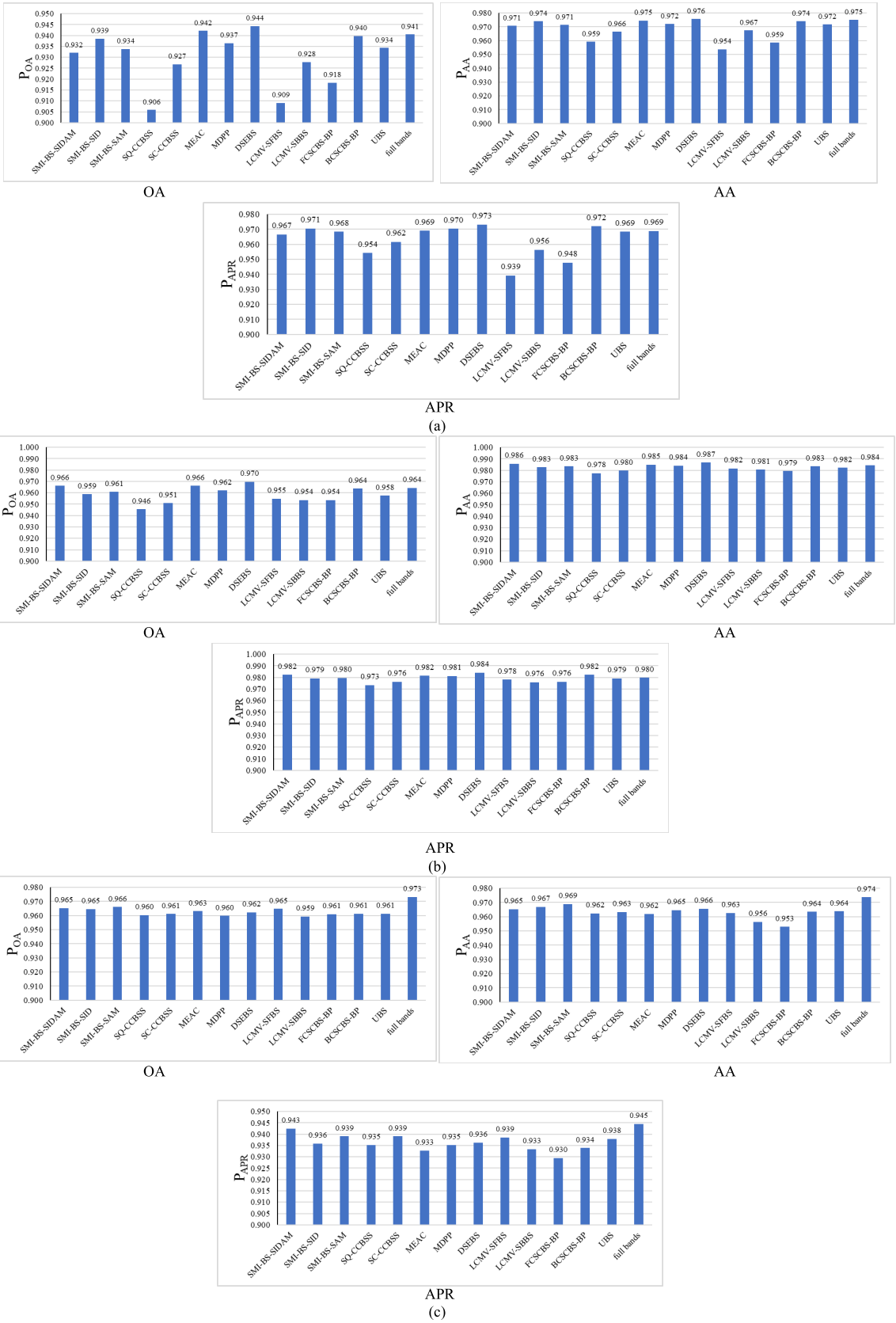


Fig. 10. Histograms of P_{OA}, P_{AA}, and P_{APR} produced by EPF-B-g, IEPF-B-g, and ICEM for Salinas. (a) EPF-B-g. (b) IEPF-B-g. (c) ICEM.

B. Salinas

The same Purdue data experiments were also performed for Salinas in Fig. 4. Unlike the Purdue Indian Pines which has imbalanced classes, Salinas scene has 16 classes, all of which are relatively large shown in Table II. Tables XII–XIV tabulate the results of P_{AA}, P_{OA}, and P_{APR} produced by EPF-B-g,

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TABLE XV

P_{OA}, P_{APR}, AND P_{AA} PRODUCED BY EPF-B-g USING BANDS SELECTED IN TABLE VIII FOR THE UNIVERSITY OF PAVIA

class	SMI-BS-SIDAM	SMI-BS-SID	SMI-BS-SAM	SQ-CCBSS	SC-CCBSS	MEAC	MDPP	DSEBS	LCMV-SFBS	LCMV-SBBS	FCSCBS-BP	BCSCBS-BP	UBS	full bands
1	0.9269	0.9313	0.9289	0.8731	0.8406	0.9003	0.9256	0.9141	0.9307	0.8784	0.8556	0.8739	0.9474	0.9629
2	0.8569	0.8945	0.8963	0.8386	0.7162	0.7524	0.8709	0.8755	0.8634	0.9151	0.6103	0.9325	0.9264	0.9732
3	0.9435	0.9356	0.9320	0.9263	0.6116	0.9551	0.9456	0.9406	0.9519	0.9235	0.6312	0.9111	0.9607	0.9463
4	0.9248	0.9310	0.9332	0.9567	0.9630	0.9154	0.9409	0.9386	0.9440	0.9458	0.7922	0.9466	0.9374	0.9412
5	0.9992	0.9995	0.9990	0.9990	0.9999	0.9991	0.9997	0.9993	0.9994	0.9984	0.9948	1.0000	0.9996	0.9990
6	0.9740	0.9573	0.9586	0.9764	0.9399	0.9257	0.9783	0.9900	0.9696	0.9944	0.5426	0.9938	0.9737	0.9966
7	0.9962	0.9981	0.9971	0.9969	0.9990	0.9989	0.9983	0.9983	0.9991	0.9963	0.9718	0.9998	0.9975	0.9992
8	0.9768	0.9734	0.9723	0.8774	0.8151	0.9655	0.9565	0.9524	0.9556	0.9371	0.7490	0.8586	0.9562	0.9706
9	1.0000	1.0000	0.9999	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9739	1.0000	1.0000	0.9999
P_{OA}	0.9129	0.9279	0.9283	0.8898	0.8068	0.8566	0.9189	0.9197	0.9159	0.9302	0.6977	0.9299	0.9464	**0.9727**
P_{APR}	0.9174	0.9185	0.9191	0.8832	0.8049	0.8747	0.9202	0.9090	0.9165	0.9079	0.7171	0.9025	0.9445	**0.9656**
P_{AA}	0.9554	0.9579	0.9575	0.9382	0.8761	0.9347	0.9573	0.9565	0.9571	0.9543	0.7913	0.9462	0.9665	**0.9765**

TABLE XVI

P_{OA}, P_{APR}, AND P_{AA} PRODUCED BY IEPF-B-g USING BANDS SELECTED IN TABLE VIII FOR THE UNIVERSITY OF PAVIA

class	SMI-BS-SIDAM	SMI-BS-SID	SMI-BS-SAM	SQ-CCBSS	SC-CCBSS	MEAC	MDPP	DSEBS	LCMV-SFBS	LCMV-SBBS	FCSCBS-BP	BCSCBS-BP	UBS	full bands
1	0.9805	0.9796	0.9752	0.9791	0.9518	0.9806	0.9787	0.9795	0.9829	0.9757	0.9492	0.9774	0.9871	0.9851
2	0.9531	0.9622	0.9630	0.9520	0.9290	0.9253	0.9642	0.9676	0.9479	0.9588	0.9071	0.9695	0.9539	0.9764
3	0.9909	0.9899	0.9859	0.9745	0.9395	0.9871	0.9895	0.9784	0.9901	0.9751	0.9469	0.9685	0.9889	0.9897
4	0.9916	0.9927	0.9941	0.9794	0.9649	0.9867	0.9913	0.9891	0.9920	0.9852	0.9480	0.9820	0.9946	0.9884
5	0.9943	0.9954	0.9923	0.9982	0.9958	0.9978	0.9994	0.9940	0.9992	0.9993	0.9888	1.0000	0.9961	0.9978
6	0.9768	0.9886	0.9807	0.9683	0.9505	0.9592	0.9852	0.9856	0.9831	0.9868	0.9326	0.9955	0.9856	0.9972
7	0.9978	0.9973	0.9975	0.9962	0.9925	0.9986	0.9980	0.9971	0.9998	0.9971	0.9889	0.9995	0.9969	0.9997
8	0.9863	0.9795	0.9753	0.9672	0.9418	0.9830	0.9899	0.9793	0.9829	0.9758	0.9564	0.9567	0.9913	0.9813
9	0.9976	0.9977	0.9954	0.9981	0.9968	0.9975	0.9993	0.9979	0.9993	0.9955	0.9695	0.9998	0.9964	0.9997
P_{OA}	0.9713	0.9759	0.9741	0.9663	0.9448	0.9564	0.9772	0.9770	0.9701	0.9722	0.9322	0.9761	0.9742	**0.9840**
P_{APR}	0.9690	0.9697	0.9696	0.9643	0.9467	0.9574	0.9735	0.9738	0.9682	0.9675	0.9358	0.9676	0.9733	**0.9784**
P_{AA}	0.9854	0.9870	0.9844	0.9792	0.9625	0.9795	0.9884	0.9854	0.9864	0.9833	0.9541	0.9832	0.9879	**0.9906**

TABLE XVII

P_{OA}, P_{APR}, AND P_{AA} PRODUCED BY ICEM USING BANDS SELECTED IN TABLE VIII FOR THE UNIVERSITY OF PAVIA

class	SMI-BS-SIDAM	SMI-BS-SID	SMI-BS-SAM	SQ-CCBSS	SC-CCBSS	MEAC	MDPP	DSEBS	LCMV-SFBS	LCMV-SBBS	FCSCBS-BP	BCSCBS-BP	UBS	full bands
1	0.8119	0.7925	0.7949	0.8157	0.7605	0.8264	0.7967	0.8041	0.8006	0.7955	0.7842	0.8270	0.8056	0.8398
2	0.9129	0.9220	0.9290	0.9248	0.9274	0.9165	0.9256	0.9233	0.9151	0.9222	0.9103	0.9182	0.9053	0.9306
3	0.9232	0.9247	0.9185	0.9166	0.9076	0.8947	0.9276	0.9300	0.9228	0.9119	0.9057	0.9309	0.9042	0.9057
4	0.7826	0.8117	0.8267	0.7859	0.7918	0.7856	0.8019	0.7614	0.7885	0.7490	0.6629	0.7790	0.7941	0.7585
5	0.9895	0.9896	0.9874	0.9799	0.9866	0.9866	0.9933	0.9911	0.9874	0.9822	0.9911	0.9948	0.9933	0.9836
6	0.9566	0.9789	0.9763	0.9513	0.9857	0.9658	0.9811	0.9642	0.9565	0.9708	0.9807	0.9775	0.9604	0.9513
7	0.9631	0.9549	0.9594	0.9744	0.9835	0.9669	0.9519	0.9714	0.9549	0.9857	0.9699	0.9820	0.9534	0.9639
8	0.8536	0.8294	0.8167	0.8528	0.8093	0.8265	0.8091	0.8121	0.8292	0.8368	0.7985	0.8436	0.8324	0.8666
9	0.8416	0.8490	0.8363	0.8501	0.8532	0.8184	0.8321	0.8574	0.8458	0.8596	0.8184	0.8606	0.8395	0.8532
P_{OA}	0.8909	0.8944	0.8970	0.8961	0.8895	0.8917	0.8942	0.8909	0.8881	0.8905	0.8738	0.8984	0.8848	**0.9009**
P_{APR}	0.6794	0.6683	0.6630	0.6810	0.6961	0.6801	0.6770	0.6774	0.6707	0.6894	0.6770	0.6983	0.6715	**0.6927**
P_{AA}	0.8928	0.8948	0.8939	0.8946	0.8895	0.8875	0.8910	0.8906	0.8890	0.8904	0.8691	0.9015	0.8876	**0.8948**

IEPF-B-g, and ICEM using the bands selected in Table VII, respectively, where the best results boldfaced by red were produced by EPF-G-g and IEPF-B-g using DSEBS. In addition, the results boldfaced by black which were better than that produced by using full bands were MEAC using EPF-B-g and SMI-BS-SIDAM and MEAC using IEPF-B-g. Table XIV also demonstrated that the results produced by ICEM using various BS methods perform slightly worse than their counterparts using full bands by 1% difference in all three classification accuracies. The conclusion was little different from that drawn from the Purdue data where ICEM actually performed better than using full bands for small classes. Nevertheless, the results in Tables XII–XIV demonstrated that SMI-BS generally performed better than most test BS methods.

Like Figs. 9 and 10 also plots the histograms of P_{OA}, P_{AA}, and P_{APR} produced by the three classifiers, EPF-B-g, IEPF-B-g, and ICEM in Tables XII–XIV for Salinas to present better visual inspection of the comparative results in Tables XII–XIV. With the same conclusion drawn from the Purdue data, the SMI-BS methods also performed consistently well for all the three classifiers compared with other BS methods.

C. University of Pavia

Like Salinas University of Pavia also has large classes with only one class less than 1000 data samples. It is different from the Purdue data with 1/4 of classes having less than 100 data samples. On the other hand, it is also different from Salinas in the sense that the University of Pavia has more complicated BKG than Salinas. Because of its image characteristics, the University of Pavia provides interesting results. For comparison, the same experiments conducted for the Purdue data and Salinas were also performed for the University of Pavia in Fig. 5. Tables XV–XVII tabulate the results of P_{AA}, P_{OA}, and P_{APR} produced by EPF-B-g, IEPF-B-g, and ICEM using the bands selected in Table VIII, respectively, where the best results boldfaced by red were produced by using full bands. Interestingly, none of BS methods could do better than full bands. This is mainly due to its complicated BKG which requires more spectral

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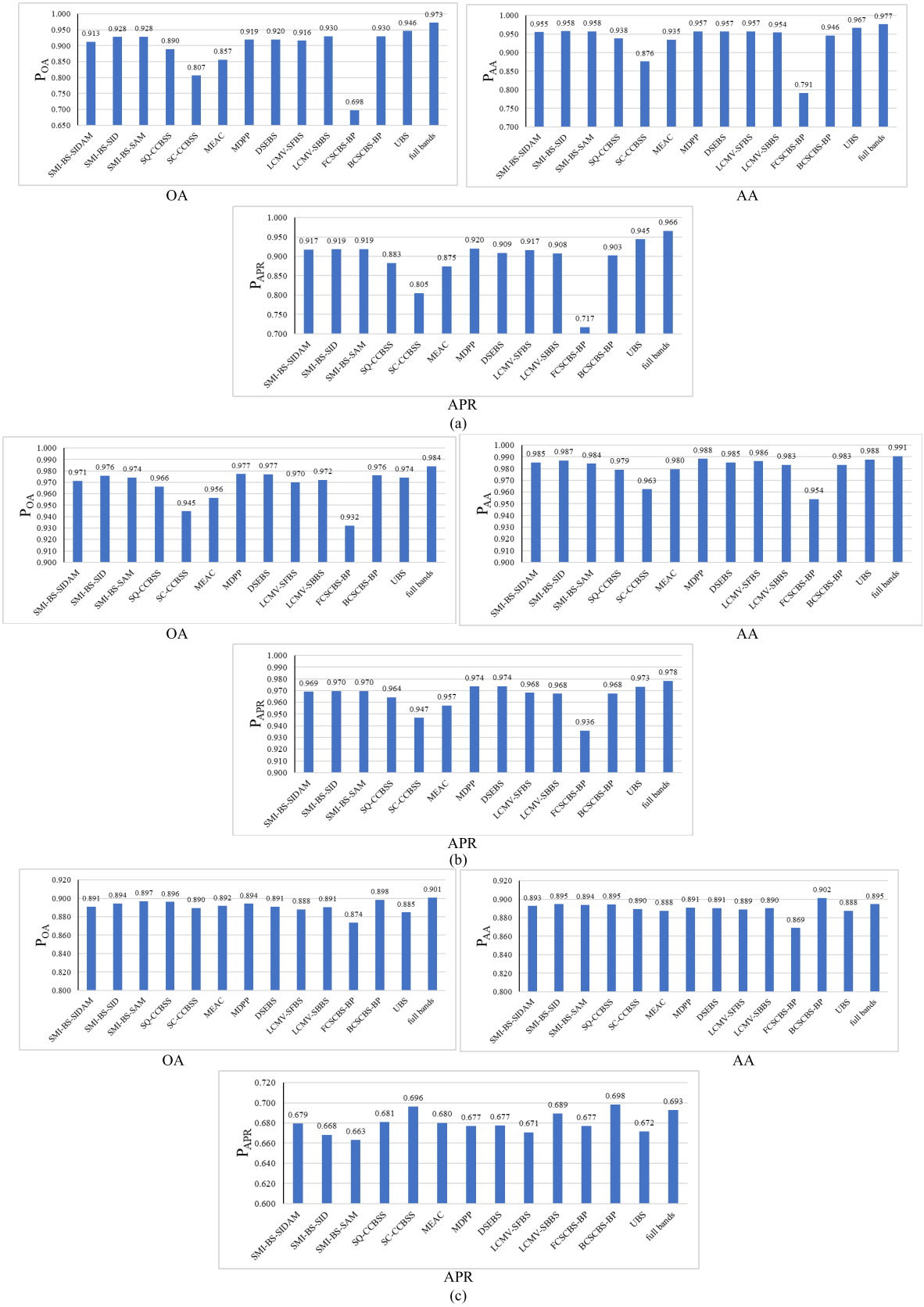


Fig. 11. Histograms of P_{OA}, P_{AA}, and P_{APR} produced by EPF-B-g, IEPF-B-g, and ICEM for the University of Pavia. (a) EPF-B-g. (b) IEPF-B-g. (c) ICEM.

information to classify edge data samples between class boundaries and BKG. As shown in Table XV, there were 4%–6% differences resulting from using selected bands when the used classifier was EPF-B-g. Such gaps were immediately closed by using the classifier IEPF-B-g within 1%, as shown in Table XVI. As for ICEM, the gaps between using full

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TABLE XVIII
COMPUTING TIME OF SMI-BS IN SECONDS FOR PURDUE DATA

Time complexity of SMI-BS in seconds for Purdue			
	SID	SAM	SIDAM
band channel construction	32	4	44
Blahut's algorithm	3	15	4
total time	36	19	48

TABLE XIX
COMPUTING TIME OF SMI-BS IN SECONDS FOR SALINAS

Time complexity of SMI-BS in seconds for Salinas			
	SID	SAM	SIDAM
band channel construction	140	11	182
Blahut's algorithm	9	22	11
total time	149	33	193

TABLE XX
COMPUTING TIME OF SMI-BS IN SECONDS FOR THE UNIVERSITY OF PAVIA

Time complexity of SMI-BS in seconds for U. of Pavia			
	SID	SAM	SIDAM
band channel construction	85	18	128
Blahut's algorithm	7	4	7
total time	92	22	135

bands and BS-selected bands were within 0.5%–1% as can be seen from Table XVII. Nonetheless, among all the test BS methods, SMI-BS generally performed better than other methods. These results also demonstrated one important fact that the used classifier is a key element to close the gaps between using full bands and bands selected by BS methods.

In analogy with Figs. 9–11 also plots the histograms of P_{OA}, P_{AA}, and P_{APR} produced by the three classifiers, EPF-B-g, IEPF-B-g, and ICEM in Tables XV–XVII for the University of Pavia to provide better visual inspection of the comparative results in Tables XV–XVII. Once again, the same conclusions drawn from the Purdue data and Salinas are applied to the University of Pavia where the SMI-BS methods also performed consistently well for all the three classifiers compared with other BS methods.

Two final comments are worthwhile. Overall speaking, SMI-BS generally performed better than other test BS methods. Specifically, for the Purdue data experiments, SMI-BS demonstrated its capability of dealing with the imbalance class issue. Compared with SQ-CCBSS and SC-CCBSS which are also information theoretical BS algorithms, the experimental results showed that SMI-BS performed better than SQ-CCBSS and SC-CCBSS. This is because SQ-CCBSS and SC-CCBSS did not really run through all possible band subsets.

IX. COMPUTING TIME COMPLEXITY ANALYSIS

According to SMI-BS, its computational complexity results from three processes, one-time construction of a hyperspectral band channel, one-time implementation of Blahut's algorithm, and prominent peak finding. Since the process of generating SMI and finding prominent peaks only took less than a second, its computing time is negligible. Tables XVIII–XX only tabulate the processing times of the hyperspectral band channel construction by SID, SAM, and SIDAM and Blahut's algorithm where a computer environment was specified by an Intel Core i7-9750H CPU 2.6 GHz with RAM: 16 G.

As we can see from Tables XVIII–XX, the major computing time is contributed by constructing a hyperspectral band channel by (5), (7), and (8). Also, the results showed that SAM required the least amount of computing time, while SIDAM required the largest amount of computing time. It makes sense since SIDAM requires an additional calculation by (8).

X. CONCLUSION

This article presents a new information theoretic approach to BS which is based on SMI. For each hyperspectral image, it constructs a hyperspectral band channel with both the band channel input space and the band channel output space specified by a full band set $\mathbf{\Omega}$ and the channel transition probabilities between input and output bands characterized by a BD measure. Furthermore, SMI is defined by $I(\mathbf{b};\mathbf{\Omega})$ as the MI between an individual band $\mathbf{b}$ and the band channel output space specified by $\mathbf{\Omega}$. Since $I(\mathbf{b};\mathbf{\Omega})$ measures the MI between two bands, $\mathbf{b}$ and $\tilde{\mathbf{b}}$, averaged over all $\tilde{\mathbf{b}} \in \mathbf{\Omega}$, it is the information of a selected band $\mathbf{b}$ to represent the remaining uncertainty after it learns from the information provided all other bands $\tilde{\mathbf{b}} \in \mathbf{\Omega}$. That is, the higher value is $I(\mathbf{b};\mathbf{\Omega})$, the more uncertainty is the band $\mathbf{b}$, thus the more significant is the band $\mathbf{b}$. With this interpretation, $I(\mathbf{b};\mathbf{\Omega})$ turns out to be a very effective criterion for BS to select bands in accordance with their uncertainties measured by $I(\mathbf{b};\mathbf{\Omega})$. Like any BP criterion interband correlation is a serious problem. To resolve this issue, two approaches are generally adopted. One commonly used method is to calculate BD among all bands to remove redundant bands, but this needs judicious selection of a threshold. The other is BSS which selects band subsets as a band package, or band clustering which selects a representative band from each band cluster to take care of interband correlation. Unfortunately, it requires knowing the size of a band subset to be selected and the number of clusters to be grouped. In addition, it needs to run through all possible band subsets or band clusters. Practically speaking, this is impossible. SMI-BS introduces a new concept of prominent peaks to select decorrelated bands that have the local maxima of an SMI curve instead of only selecting bands with high SMI to avoid adjacent bands being selected as to reduce correlation between selected bands.

The SMI-BS proposed in this article has several potential research directions. One is to use SMI as a class feature to measure class uncertainly and entropy in HSIC. A second direction is to take care of band correlation issue via prominent peaks to make BP criteria more effective. A third one is to use the concept of prominent peaks to determine the number of bands needed to be selected instead of using VD as used in this article. Finally, SMI can be used in conjunction with autoencoder [41] for BS.

APPENDIX

In the algorithm for finding prominent peaks described in Section V, a MATLAB method available on its website, www.mathworks.com/help/signal/ref/findpeaks.html, is used to determine the prominent peaks of an SMI curve and can be summarized as follows.

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It is the function of `pks = findpeaks(data, 'MinPeakProminence', Value)` returns a vector with the local maxima (peaks) that have a relative importance of at least the value of `'MinPeakProminence'` of the input signal vector, `data`. A *local peak* is a data sample that is either larger than its two neighboring samples or is equal to `Inf`. Non-`Inf` signal endpoints are excluded. If a peak is flat, the function returns only the point with the lowest index. Most importantly, it defines the concept of prominence of a peak to measure how much the peak stands out due to its intrinsic height and its location relative to other peaks. A low isolated peak can be more prominent than one that is higher but is an otherwise unremarkable member of a tall range.

To measure the prominence of a peak:

1) Place a marker on the peak.
2) Extend a horizontal line from the peak to the left and right until the line does one of the following:
   a) crosses the signal because there is a higher peak
   b) reaches the left or right end of the signal
3) Find the minimum of the signal in each of the two intervals defined in Step 2. This point is either a valley or one of the signal endpoints.
4) The higher of the two interval minima specifies the reference level. The height of the peak above this level is its prominence.

It should be noted that `findpeaks` make no assumption about the behavior of the signal beyond its endpoints, whatever their height. This is reflected in steps 2 and 4 and often affects the value of the reference level.

REFERENCES

[1] T. M. Cover and J. A. Thomas, *Elements of Information Theory*. Hoboken, NJ, USA: Wiley, 2012.

[2] R. G. Gallager, *Information Theory and Reliable Communication*. Hoboken, NJ, USA: Wiley, 1968.

[3] R. E. Blahut, *Principles and Practice of Information Theory*. Reading, MA, USA: Addison-Wesley, 1987.

[4] C.-I Chang, "An information-theoretic approach to spectral variability, similarity, and discrimination for hyperspectral image analysis," *IEEE Trans. Inf. Theory*, vol. 46, no. 5, pp. 1927–1932, Aug. 2000.

[5] Y. Du, C.-I Chang, H. Ren, C.-C. Chang, and J. O. Jensen, "New hyperspectral discrimination measure for spectral characterisation," *Opt. Eng.*, vol. 43, no. 8, pp. 1777–1786, Aug. 2004.

[6] C.-I Chang, L.-C. Lee, B. Xue, M. Song, and J. Chen, "Channel capacity approach to hyperspectral band subset selection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 10, pp. 4630–4644, Oct. 2017.

[7] C.-I Chang, "Spectral inter-band discrimination capacity of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1746–1766, Mar. 2018.

[8] C.-I Chang, K. Y. Ma, C.-C. Liang, Y.-M. Kuo, and S. Chen, "Iterative random training sampling spectral-spatial classification for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3986–4007, Jul. 2020.

[9] M. Song, X. Shang, Y. Wang, C. Yu, and C.-I Chang, "Class-information based band selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8394–8416, Nov. 2019.

[10] R. M. Fano, *Transmission of Information*. Cambridge, MA, USA: MIT Press, 1961.

[11] R. Blahut, "Computation of channel capacity and rate-distortion functions," *IEEE Trans. Inf. Theory*, vol. IT-18, no. 4, pp. 460–473, Jul. 1972.

[12] C.-I Chang, Q. Du, T.-L. Sun, and M. L. G. Althouse, "A joint band prioritization and band-decorrelation approach to band selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 6, pp. 2631–2641, Nov. 1999.

[13] H. Yang, Q. Du, H. Su, and Y. Sheng, "An efficient method for supervised hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 1, pp. 138–142, Jan. 2011.

[14] Q. Wang, Q. Li, and X. Li, "Hyperspectral band selection via adaptive subspace partition strategy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4940–4950, Dec. 2019.

[15] Q. Wang, Q. Li, and X. Li, "A fast neighborhood grouping method for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 608–619, Jul. 2020.

[16] M. Baisantry, A. K. Sao, and D. Shukla, "Band selection using combined divergence-correlation index and sparse loadings representation for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, no. 8, pp. 4630–4644, Aug. 2020.

[17] C.-I Chang, *Hyperspectral Data Processing: Signal Processing Algorithm Design and Analysis*. Hoboken, NJ, USA: Wiley, 2013.

[18] C. C. Wu, S. Chu, and C.-I Chang, "Sequential N-FINDR algorithm," in *Proc. SPIE Conf. Imag. Spectrometry XIII*, San Diego, CA, USA, Aug. 2008, Art. no. 70860C, doi: 10.1117/12.795262.

[19] L. Wang and C.-I Chang, "Multiple band selection for anomaly detection in hyperspectral imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 7022–7025.

[20] L. Wang *et al.*, "Band subset selection for anomaly detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 4887–4898, Sep. 2017.

[21] H.-C. Li, C.-I Chang, L. Wang, and Y. Li, "Constrained multiple band selection for hyperspectral imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Beijing, China, Jul. 2016, pp. 6149–6152.

[22] W. Xiong, C.-I Chang, C.-C. Wu, K. Kalpakis, and H. M. Chen, "Fast algorithms to implement N-FINDR for hyperspectral endmember extraction," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 4, no. 3, pp. 545–564, Sep. 2011.

[23] C.-I Chang, "Maximum simplex volume-based endmember extraction algorithms," U.S. Patent 8 417 748 B2, Apr. 9, 2013.

[24] H. Su, Q. Du, G. Chen, and P. Du, "Optimized hyperspectral band selection using particle swarm optimization," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2659–2670, Jun. 2014.

[25] H. Su, B. Yang, and Q. Du, "Hyperspectral band selection using improved firefly algorithm," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 1, pp. 68–72, Jan. 2016.

[26] Y. Yuan, G. Zhu, and Q. Wang, "Hyperspectral band selection by multitask sparisty pursuit," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 631–644, Feb. 2015.

[27] Y. Yuan, X. Zheng, and X. Lu, "Discovering diverse subset for unsupervised hyperspectral band selection," *IEEE Trans. Image Process.*, vol. 26, no. 1, pp. 51–64, Jan. 2017.

[28] G. Zhu, Y. Huang, J. Lei, Z. Bi, and F. Xu, "Unsupervised hyperspectral band selection by dominant set extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 227–239, Jan. 2016.

[29] C.-I Chang, *Hyperspectral Imaging: Spectral Techniques for Detection and Classification*. Norwell, MA, USA: Kluwer, 2003.

[30] C.-I Chang and Q. Du, "Estimation of number of spectrally distinct signal sources in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 3, pp. 608–619, Mar. 2004.

[31] C.-I Chang, "A review of virtual dimensionality for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1285–1305, Apr. 2018.

[32] X. Kang, S. Li, and J. A. Benediktsson, "Spectral-spatial hyperspectral image classification with edge-preserving filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2666–2677, May 2014.

[33] C.-I Chang, "A statistical detection theory approach to hyperspectral image classification: Performance analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2057–2074, Apr. 2019.

[34] M. Song, X. Shang, and C.-I Chang, "3-D receiver operating characteristic analysis for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, early access, Apr. 29, 2020, doi: 10.1109/TGRS.2020.2987137.

[35] C.-I Chang, W. Xiong, and C.-H. Wen, "A theory of high-order statistics-based virtual dimensionality for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 188–208, Jan. 2014.

[36] J. C. Harsanyi, W. Farrand, and C.-I Chang, "Detection of subpixel spectral signatures in hyperspectral image sequences," in *Proc. Amer. Congr. Surveying Mapping (ACSM)/Amer. Soc. Photogramm. Remote Sens. (ASPRS) Annu. Conv. Expo.*, Baltimore, MD, USA, vol. 1, 1994, pp. 236–247.

[37] C. Yu, M. Song, Y. Wang, and C.-I Chang, "Class signature-constrained background-suppressed approach to band selection for classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 14–31, Jan. 2019.

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[38] C. Yu, M. Song, and C.-I Chang, "Band subset selection for hyperspectral image classification," *Remote Sens.*, vol. 10, no. 1, p. 113, Jan. 2018, doi: 10.3390/rs10010113.
[39] S. Zhong, C.-I Chang, and Y. Zhang, "Iterative edge preserving filtering approach to hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 90–94, Jan. 2019.
[40] B. Xue *et al.*, "A subpixel target detection approach to hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5093–5114, Sep. 2017.
[41] M. A. Kramer, "Nonlinear principal component anlayis using autoassociative neural networks," *AIChE J.*, vol. 37, no. 2, pp. 233–243, 1991.



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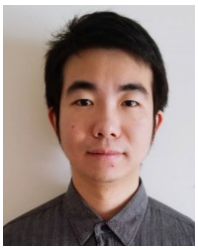
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